Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends*

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Abstract

Large incumbent firms have grown disproportionately relative to smaller firms in many advanced economies, leading to rising firm-size dispersion. This paper provides a decomposition of this increase into three distinct margins of firm dynamics: the firm size-age relationship, the firm-age distribution, and size heterogeneity conditional on age. Using a heterogeneous-firms model disciplined by Swedish administrative data, I quantify the contribution of these margins to the rise in firm-size dispersion. The results show that all three margins have contributed significantly to the increase in dispersion. The underlying structural changes imply a welfare loss of 4.95% in permanent consumption units.

Keywords: Firm dynamics, Long-run macroeconomic trends, Administrative data

JEL codes: D22, E24, O31, O47, O50

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1 Introduction

Over recent decades, advanced economies have exhibited a striking macroeconomic development: large incumbent firms have grown disproportionately relative to smaller firms, leading to a pronounced rise in firm-size dispersion (Autor et al., 2020; Akcigit and Ates, 2021).[1] This paper shows that this aggregate pattern can emerge from markedly different underlying firm dynamics. I provide a decomposition of firm-size dispersion into distinct, observable margins of firm dynamics, allowing these forces to be separately quantified. I implement this decomposition using comprehensive Swedish administrative data to measure the contribution of each margin to the rise in firm-size dispersion.

The paper begins with a model-free decomposition that relates cross-sectional firm-size dispersion to different margins of firm dynamics. Dispersion rises when the average size of surviving firms increases more steeply with age, when the firm-age distribution shifts toward older (and therefore larger) firms, or when size heterogeneity conditional on age expands. The decomposition shows that dispersion can rise for three structurally distinct reasons, which are observationally equivalent at the macroeconomic level.

Distinguishing between these margins is important for at least two reasons. First, the decomposition is informative about the underlying cause of rising firm-size dispersion. A steepening of the firm size-age relationship—corresponding to the first component of the decomposition—may reflect changes in firm expansion costs (Cao et al., 2022), while a shift in the firm-age distribution or rising size dispersion conditional on age may point to, increasing entry barriers (Davis, 2017; Gutiérrez et al., 2021) or rising productivity differences between firms (Aghion et al., 2023), respectively. Which of these forces drives the increase in firm-size dispersion ultimately matters for welfare. Second, the decomposition has direct policy implications. If, for example, the rise in dispersion reflects a shift in the firm-age distribution, policies that stimulate entry may be effective.

Using high-quality administrative data from Sweden, I document a marked rise in cross-sectional firm-size dispersion. This increase is accompanied by three developments at the micro level. First, consistent with trends in the United States (Hopenhayn et al., 2022; Karahan et al., 2024), the share of young firms—measured by the entry rate—has declined from 15% in 1997 to 11% in 2017. Second, in contrast to the U.S. evidence, the firm size-age relationship has steepened over time in Sweden. For example, in the 1990s, average real value added of surviving firms at age eight was 49% higher than the average size of entrants; by the 2010s, the corresponding figure had risen to 67%. Third, size dispersion conditional on age has increased among older firms.

How much does each margin contribute to the total rise in firm-size dispersion? Implementing the decomposition directly would require observing the ages of all firms. However, the ages of older firms cannot be fully observed in administrative data.[2] I therefore develop a

[1]These trends are further documented in, among others, Grullon et al. (2019); Decker et al. (2016, 2020); Gourio et al. (2014); Chen et al. (2023).

[2]Firm age is typically top-coded in administrative data, as firm age is usually measured from the first

structural model of firm dynamics that allows me to recover the implied size profiles and age distribution, while disciplining the model with the observed counterparts. The model also allows me to quantify the welfare consequences of the rise in firm-size dispersion.

At its core, the model is an endogenous-growth framework of creative destruction in the spirit of the quality-ladder models of Aghion and Howitt (1992) and Klette and Kortum (2004). Incumbent firms and potential entrants expand into new markets by improving the quality of competitors' products through innovation (expansion R&D). In this environment, expansion costs discipline the firm size-age relationship, while entry costs govern firm entry and hence shape the firm-age distribution. To capture the third margin of the dispersion decomposition, I extend the Klette-Kortum model by introducing ex-ante heterogeneity in firm productivity: entrants exogenously differ in their production efficiency, with some systematically more efficient than others. In equilibrium, more productive firms expand faster into new product markets than less productive ones, generating size dispersion conditional on firm age. As a result, expansion costs, entry costs, and ex-ante productivity differences map into the three components of the dispersion decomposition.

I estimate the model on separate balanced growth paths and decompose the change in total firm-size dispersion across them. For the initial balanced growth path, I target empirical moments from the late 1990s, including the firm size-age relationship, the firm entry rate (which governs the age distribution), total firm-size dispersion (which disciplines dispersion conditional on age as a residual), and aggregate productivity growth. The subsequent balanced growth path targets the corresponding moments for the 2010s, in particular the steepening of the size-age relationship, the four-percentage-point decline in the firm entry rate, the rise in total firm-size dispersion, and a 0.4 percentage point fall in the aggregate productivity growth rate as observed for the Swedish economy. The estimated changes in model primitives—among them a decline in firm expansion costs and an increase in ex-ante productivity heterogeneity—jointly account for the changes in the targeted moments.

Having estimated the separate balanced growth paths, I decompose the change in total firm-size dispersion into its three components. The results indicate that 34.4% of the overall increase is attributable to the steepening of the firm size-age relationship, 28.5% to the aging of firms, and 26.2% to the rise in dispersion conditional on age.[3] Although the steepening of the size-age relationship accounts for the largest share, all three margins contribute significantly to the increase in dispersion. In particular, while size dispersion conditional on age has received relatively little attention in the literature, it accounts for an almost equal share of the observed rise in cross-sectional firm-size dispersion in the Swedish data as changes in the age distribution.

I further use the model to extend the analysis in two directions. First, I solve for the transition path between the balanced growth paths to quantify the implications for welfare.

year a firm appears in the data. Even when business registries record incorporation dates, founding years are not observed for firms that pre-date the registry's establishment, so firm ages remain effectively top-coded.

[3] These shares do not sum to 100% because the decomposition is not additive.

Welfare falls by 4.95% in permanent consumption units, implying that the structural changes underlying the rise in firm-size dispersion have, on net, reduced welfare. Second, I decompose the steepening of the firm size-age relationship into changes in average size conditional on age within each productivity type and changes in the type composition of surviving firms. The model implies that more productive firms not only have become larger conditional on age, but also account for a higher share of firms at older ages, so that selection effects contribute to the steeper size-age relationship.

Further Literature. This paper relates to a growing literature that studies the drivers of recent macroeconomic trends, including rising firm-size dispersion, declining entry, and slowing productivity growth. Proposed mechanisms include falling expansion costs (Cao et al., 2022), increasing entry barriers (Davis, 2017; Gutiérrez et al., 2021), rising productivity dispersion (Aghion et al., 2023), declining research productivity (Bloom et al., 2020; Olmstead-Rumsey, 2019) or the growing importance of intangible capital in firms' production (De Ridder, 2024; Chiavari and Goraya, 2020; Weiss, 2019). While this literature typically seeks to identify specific exogenous—often unobserved—forces behind aggregate developments, the objective here is different. Rather than isolating the exogenous driver, the paper decomposes the rise in firm-size dispersion into three endogenous and observable margins of firm dynamics. This approach clarifies which margins account for the observed changes and provides a policy-relevant mapping from aggregate outcomes to underlying firm behavior.

The paper is also related to empirical work studying firm dynamics in the United States. Hopenhayn et al. (2022) and Karahan et al. (2024) document that the firm-age distribution has shifted toward older firms, while the size-age relationship has remained relatively stable. Because these papers focus on explaining the rise in average firm size, they do not examine size dispersion conditional on age for firms of all ages. As a result, it remains unclear which margin in the decomposition drives the increase in total firm-size dispersion in the U.S.

Other studies have embedded ex-ante firm heterogeneity into models of creative destruction. Aghion et al. (2023) develop a model with ex-ante productivity differences, but abstract from firm entry and exit, which are essential for analyzing the firm size-age relationship. Ex-ante heterogeneity has also been introduced along other dimensions—for example, in innovation step sizes (Lentz and Mortensen, 2008). Heterogeneity in productivity is conceptually distinct: firms must track the entire distribution of productivity types across product markets to form markup expectations when expanding into new lines. This introduces an additional state variable—the productivity distribution—into the firm's optimization problem. By contrast, with heterogeneity in step sizes or R&D costs, firms only need to track their own type.

The remainder of the paper proceeds as follows. Section 2 introduces a simple decomposition linking different margins of firm dynamics to cross-sectional size dispersion. Section 3 documents empirical trends in firm dynamics and size dispersion in Sweden. Section 4 presents the structural model used to quantify the contributions of the different margins. Section 6 provides extensions, including the welfare analysis, and Section 7 concludes.

2 A model-free decomposition of firm-size dispersion

To guide the analysis of firm dynamics and firm-size dispersion in the next sections, I begin with a model-free decomposition that links both. Consider the law of total variance, $\text{Var}(Y) = \text{Var}(E[Y|X]) + E[\text{Var}(Y|X)]$, which decomposes the total dispersion in Y into (i) dispersion between groups defined by X and (ii) residual dispersion within those groups. Setting $Y \equiv \ln s_f$ and $X \equiv a_f$, where s_f and a_f denote firm size and age, respectively, this decomposition partitions the cross-sectional dispersion in log firm size into variation in average size across firm ages and residual heterogeneity in size conditional on age.[4] Specifically,

$$\begin{aligned}\text{Var}\left(\ln s_f\right) = &\sum_{a_f} p(a_f)\left[E[\ln s_f|a_f] - E[\ln s_f|0] - \sum_{a_f} p(a_f)\left(E[\ln s_f|a_f] - E[\ln s_f|0]\right)\right]^2 \\ &+ \sum_{a_f} p(a_f)\text{Var}(\ln s_f|a_f),\end{aligned} \tag{1}$$

where $p(a_f)$ is the share of firms of age a_f. In the first term, I added (and subtracted) the average log size of entrants, $E[\ln s_f|0]$, to express between-age variation in terms of deviations from average entry size.

Although the decomposition is specific to the variance, it more generally highlights that cross-sectional firm-size dispersion is shaped by three components: (i) the firm size-age relationship, i.e., the average size of firms alive at age $a_f \in \{1, 2, 3 \ldots\}$ relative to the average size of entrants, $E[\ln s_f|a_f] - E[\ln s_f|0]$; (ii) the age distribution of firms, $p(a_f)$; and (iii) residual size heterogeneity within age groups, $\text{Var}(\ln s_f|a_f)$. Total dispersion can increase for three reasons: the size-age relationship may steepen, firm sizes within age brackets may become more dispersed, or the age distribution may shift toward older firms, which are typically larger and exhibit greater within-age dispersion.

Equation (1) provides a natural framework for decomposing changes in cross-sectional size dispersion into these three components. Empirically, however, implementing this decomposition using administrative data that cover the universe of firms is challenging because firm age is typically measured from the first year of observation, implying that firms already active at the beginning of the data have top-coded ages. Even when business registries record incorporation dates, founding years are not observed for firms that pre-date the registry, leaving ages of the oldest and largest firms effectively truncated. I therefore begin by documenting trends in firm dynamics for observable age groups. In subsequent sections, I employ a structural model that recovers firms' entire size trajectories and the full age distribution, allowing for an exact decomposition of overall size dispersion into its three components.

[4] While I use the variance to derive a tractable analytical expression, the underlying intuition extends to alternative dispersion measures.

3 Trends in the Swedish economy

Before documenting the empirical trends, I describe the data first.

3.1 Data

Data is provided by Statistics Sweden, the official statistical agency in Sweden. The main data set is *Företagens Ekonomi*, which covers information from balance sheets and profit and loss statements for the universe of Swedish firms at an annual frequency covering the period 1997-2017. The empirical analysis is based on the resulting unbalanced panel of firms. The data contains two measures of firm size that I use throughout the paper: value added and the wage bill. For a detailed description of the variables, see Section A in the Appendix. The data further contains information on the firm's industry and legal type. I restrict the analysis to firms operating in the private economy. The firm's birth year is defined as the year the first worker is associated with a firm (employed or self-employed). I obtain this information from the auxiliary data set *Registerbaserad Arbetsmarknadsstatistik*, containing the universe of employer-employee matches. Firms entering late during the year compare small, on average, to firms entering early. To prevent the timing of entry from affecting the average size of entrants within a year, I define the latter as firms aged zero or one. I further restrict the analysis to firms that eventually hire at least one employee, which filters out never-employing firms.

Starting in 2004, the data includes unincorporated self-employed with negative profits. These firms were previously not recorded, leading to an inflow of new firms in that year. The firm entry rate in 2004 is hardly distinguishable from its trend in the cleaned data, suggesting that the stock of these firms is very small. Nevertheless, since I cannot differentiate between firms established in 2004 and firms entering the data in 2004 with an earlier birth year, I drop the cohort of firms entering the data in 2004. For the results of the paper, the change in the data is inconsequential, as the trends I document mostly occur before 2004. Further, the characteristics of entrants have been very stable over time, as shown below, indicating no systematic changes in the type of firms entering the data.

Table 1: Summary statistics (1997-2017)

	25th Pct.	50th Pct.	75th Pct.	Mean	SD	Obs.
*Sales**	0.5	1.4	4.4	18.7	464.8	7,374,865
*Value added**	0.2	0.6	1.7	5.1	116.4	7,374,865
Employment (full-time units)	0	1	3	6.6	107.3	7,374,865
*Wage bill**	0.002	0.24	0.87	2.45	43.4	7,374,865
*Capital stock**	0.02	0.16	0.8	6.9	228.5	7,374,865
*Intermediate Inputs**	0.2	0.5	1.6	7.3	220.8	7,374,865

Note: variables marked with * are in units of million 2017-SEK (1 SEK ≈ 0.1 US dollars). The capital stock is defined as fixed assets minus depreciation.

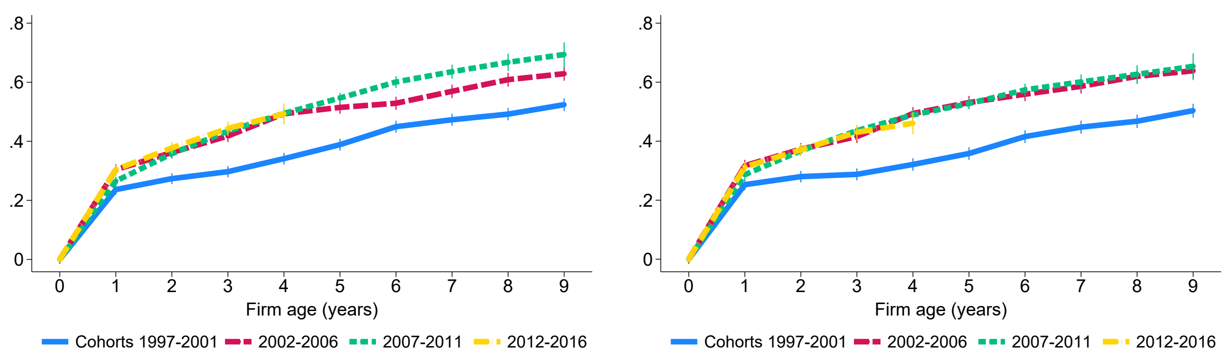
Table 1 reports distributional statistics of firm sales, value added, and production inputs in the data (1997 to 2017). Throughout the paper, nominal variables are deflated to 2017 Swedish Krona (SEK) using the GDP deflator. The median firm lists sales of roughly 1.4 million SEK (approx. 0.14 million US dollars) and employs one full-time worker. The distribution of sales, value added, and all production inputs is highly right-skewed. Mean sales (18.7 million SEK) and full-time employment (6.6) far exceed their respective 75th percentiles. In total, the data includes about 7.4 million firm-year observations. For the firm age- and size-specific empirical analysis, I filter value added at its 0.5% tails and focus on firms established in 1997 or later, which reduces the sample size to 3.5 million firm-year observations.[5] For firms established in 1997 and later, age is not truncated.

3.2 Trends in firm dynamics

This section documents changes in the size dynamics of firms. As a starting point, I nonparametrically characterize the relationship between firm size and firm age. Figure 1 plots the average size of surviving firms at each age, normalized by the average size of entrants. The relative size is presented in logs. Panel (a) uses value added to measure firm size, while Panel (b) uses the wage bill. Both panels display 95% confidence intervals around the estimated means.

Figure 1: Firm size-age relationship (by cohort)

(a) Value added (b) Wage bill



Notes: the figure shows the average value added (left) and wage bill (right) of surviving firms as a function of firm age relative to the average size of entrants in logs for different cohorts as indicated in the legend. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are shown.

As expected, the average size of surviving firms increases with age. More notably, however, the relationship between age and size has steepened over time. Figure 1 presents the size-age relationship separately for the cohorts 1997–2001, 2002–2006, 2007–2011, and 2012–2016.

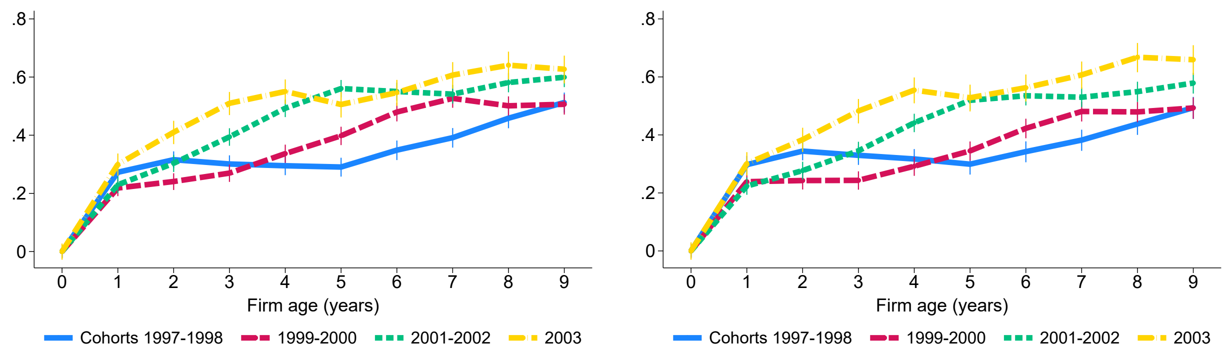
[5] I apply the 0.5% filter also to the wage bill and the capital stock, but only at the right tail. This ensures that self-employed without labor costs or firms without any physical capital are included in the analysis.

Compared to the initial cohorts (1997–2001), the relative size of surviving firms has increased for any firm age across all subsequent cohorts. For the 1997–2001 cohorts, firms that survive to age eight are on average 0.492 log points larger in terms of value added than entrants; for the 2007–2011 cohorts, the corresponding figure is 0.668 log points. A similar pattern emerges for the wage bill: at age eight, it is 0.468 log points above average entry size for the 1997–2001 cohorts, compared with 0.627 log points for the 2007–2011 cohorts.

Figure 2: Firm size-age relationship (cohorts 1997–2003)

(a) Value added

(b) Wage bill



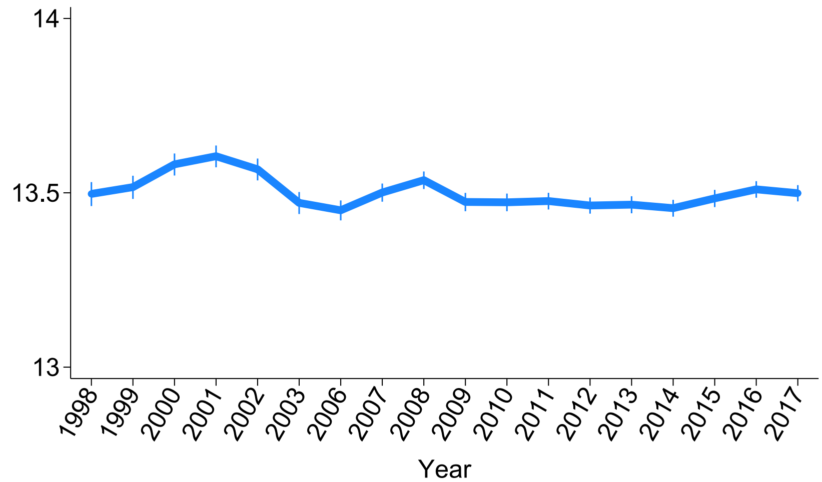
Notes: the figure shows the average value added (left) and wage bill (right) of surviving firms as a function of firm age relative to the average size of entrants in logs for different cohorts as indicated in the legend. 95% confidence bands are included.

Figure 1 indicates that the firm size-age steepening was concentrated primarily among cohorts from the late 1990s and early 2000s. To examine this pattern in greater detail, I focus specifically on these cohorts. Figure 2 presents the size-age relationship for the 1997–1998, 1999–2000, 2001–2002, and 2003 cohorts individually. The figure shows that the steepening occurred gradually for these cohorts. Compared to Figure 1, the trajectories appear less smooth because the averages are calculated over two cohorts rather than five. Nonetheless, a gradual steepening is evident for both value added and the wage bill. In fact, the trajectory of the 2003 cohort closely resembles that of the later cohorts shown in Figure 1. Note that the trajectories in Figures 1 and 2 depict the size profiles of surviving firms and therefore do not necessarily imply that firms grow faster today. Changes in the composition of surviving firms may also account for these patterns, a point to which I return below.

The firm size-age relationship has steepened over time, while the size of entrants has remained remarkably stable. Figure 3 plots the log of the average wage bill for entrant firms. Firm size at entry is highly stable, particularly over the long run. This stability suggests that there have been no systematic changes in the nature of firms entering the data over time. The years 2004 and 2005 are omitted from the graph because, as noted in the previous section, the 2004 entrant cohort is excluded, and entrants are defined as firms that are zero or one year old. For clarity, all year labels are shown.

I provide additional robustness checks in Appendix B.1. For instance, Figure A-1a displays

Figure 3: Firm size at entry



Notes: the figure shows the average firm size (wage bill) at entry in logs. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one. 95% confidence bands are included.

the relationship between firms' age and their physical capital stock across all cohorts. The trajectories are nearly identical across cohorts, indicating no systematic differences in capital accumulation over time. This suggests that the observed steepening of the size-age relationship is not driven by capital deepening. This conclusion holds even when adding measured intangible assets to the capital stock, as shown in Figure A-1b. I also display the firm-size relationship by sector in Figure A-2. The steepening is evident in both goods production and services, though it is particularly pronounced among services firms. This pattern implies that the results are unlikely to be driven by external factors, such as foreign demand for Swedish goods. Lastly, I confirm the steepening of the size-age relationship using a regression framework, controlling for cohort and narrowly defined industry fixed effects.

How have the firm-age distribution and size dispersion conditional on age evolved over time? Figure A-3 in the Appendix shows that size dispersion conditional on age remained roughly constant for younger firms, while it increased for older firms. For firms aged ten and above, a clear upward trend emerges. For example, among firms at age ten, size dispersion increased by about 0.2 log points. Assessing changes in the full age distribution is more challenging because firm age is not observed for firms established prior to 1993.[6] In the following section, I report the evolution of the entry rate, defined as the ratio of firms aged zero or one to the total number of firms. The entry rate declines over time, indicating a shrinking share of young firms in the economy.

[6]As described in Section 3.1, firm age is measured using the auxiliary data set *Registerbaserad Arbetsmarknadsstatistik*, which dates back to 1992. The main data set, *Företagens Ekonomi*, starts in 1997.

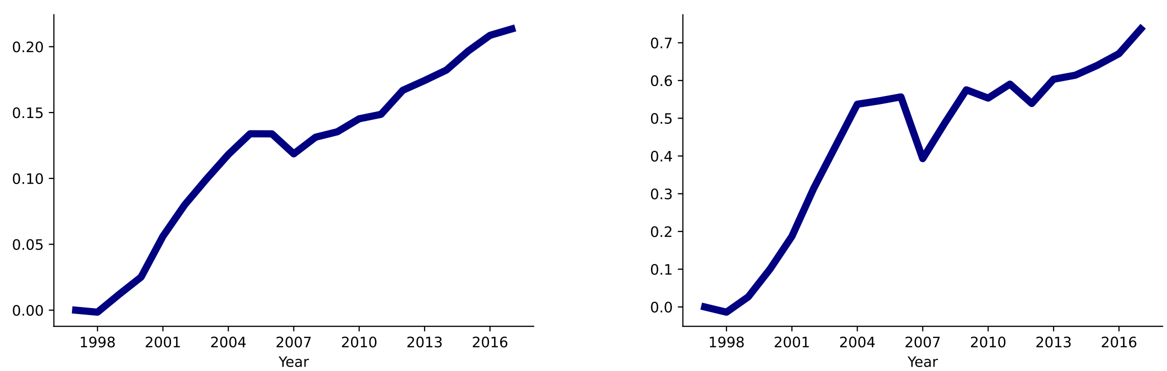
3.3 Trends in the macroeconomy

Using the same data, this section documents systematic macroeconomic trends, beginning with rising firm-size dispersion. Figure 4 shows the evolution of two cross-sectional dispersion measures for log value added: the standard deviation and the interquartile range (defined as the difference between the 75th and 25th percentiles). Both measures are calculated within two-digit industries, aggregated using industry labor costs, and normalized to zero in 1997. For the standard deviation (Figure 4a), the sample is restricted to firms with positive value added, as the logarithm is otherwise undefined. For the preferred measure—the interquartile range in Figure 4b—this restriction is not imposed, since the 25th percentile within two-digit industries is typically positive.[7]

Figure 4: Cross-sectional firm size dispersion

(a) Standard deviation

(b) Interquartile range



Notes: the left panel shows the standard deviation and the right panel the interquartile range of log value added. Both measures are computed within two-digit industries and aggregated using industry labor costs. The value in 1997 is normalized to zero.

The interquartile range reveals a pronounced increase in size dispersion over the period 1997–2017: the 75th percentile rose by 0.737 log points relative to the 25th percentile. This implies that, had the two percentiles been equal in 1997, firms at the 75th percentile would be approximately $e^{0.737} \approx 2.09$ times larger than those at the 25th percentile by 2017. Notably, a significant portion of this increase—over 0.5 log points—occurred during the early 2000s. The rise in the interquartile range is more pronounced than the increase in the standard deviation—suggesting that dispersion grew more in the center of the distribution than in the tails—though this comparison should be interpreted with caution given the restriction of the standard deviation to firms with positive value added.

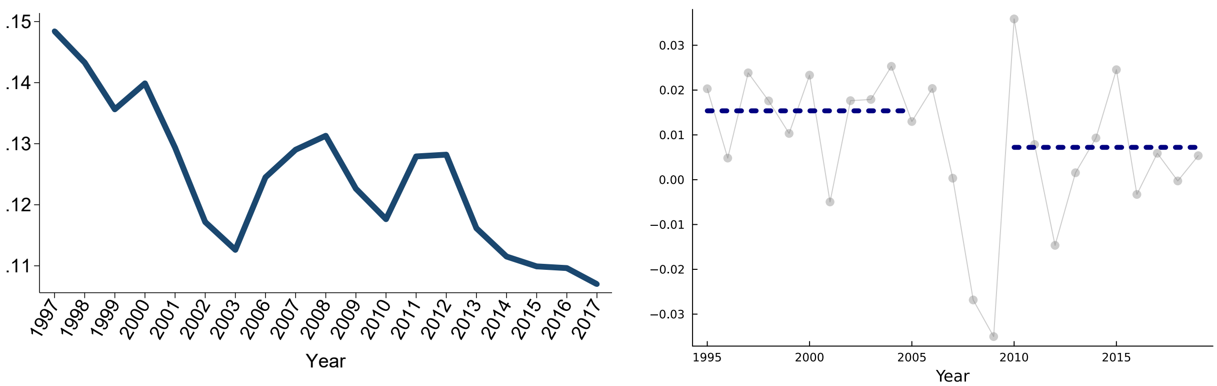
The rise in firm-size dispersion occurred alongside other macroeconomic trends that are

[7] The results are very similar when using finer, five-digit industry classifications, as shown in Appendix B.2. However, this greater granularity increases the frequency of negative values at the 25th percentile of value added, which makes the interquartile range undefined for those product groups.

common across developed economies. Figure 5a shows the evolution of the firm entry rate between 1997 and 2017. Over this period, the entry rate declined from about 15% in 1997 to 11% in 2017. Notably, most of this four-percentage-point drop took place around the turn of the millennium. Although the entry rate experienced a modest rebound following the Great Recession, it soon returned to—and even fell below—its earlier levels.[8] As before, the years 2003 and 2004 are excluded, and all year labels are shown in the figure for clarity.

Figure 5: Other macroeconomic trends

(a) Entry rate (b) Total Factor Productivity growth



Notes: the left panel shows the entry rate defined as the total number of entrants divided by the total number of firms. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one. Right panel: the scatter points in gray show annual Total Factor Productivity (TFP) growth from Federal Reserve Economic Data (FRED). The dashed blue line indicates the respective averages over 1995–2005 and 2010–2019.

Figure 5b presents annual Total Factor Productivity (TFP) growth obtained from Federal Reserve Economic Data (FRED). The series is plotted as scatter points, with the dashed blue line indicating time-averaged values. Sweden experienced strong TFP growth from the mid-1990s through the early 2000s, averaging 1.5% annually between 1995 and 2005. Following the early 2000s, growth slowed, with TFP increasing at an average annual rate of around 1.1% between 2010 and 2015. This pattern—a period of rapid growth around the turn of the millennium followed by a slowdown during the 2010s—mirrors trends observed in many advanced economies, including the U.S. (Aghion et al., 2023).

4 Model

To extend the analysis of firm dynamics beyond the observable age groups, the next section develops a structural model. The model recovers firms' complete life-cycle size profiles and the full age distribution, permitting an exact decomposition of firm-size dispersion across all ages. It also provides a framework to assess the welfare consequences of the structural changes underlying the observed trends.

[8] Engbom (2023) reports a similar decline in the firm entry rate for Sweden.

4.1 Preferences and aggregate economy

Time is continuous and indexed by t. The economy consists of a representative household that chooses the path of consumption C_t and wealth A_t to maximize lifetime utility

$$U = \int_0^{\infty} \exp(-\rho t) \ln C_t dt,$$

subject to the budget constraint $\dot{A}_t = r_t A_t + w_t L_t - C_t$. ρ denotes the discount rate, r_t the interest rate and w_t the real wage. The household supplies one unit of labor inelastically, i.e., $L_t = 1$. The optimality condition (Euler equation) for the household problem reads

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

Aggregate output is produced competitively using a Cobb-Douglas technology over a continuum of different products indexed by i (t subscripts are suppressed when convenient)

$$Y = \exp\left(\int_0^1 \ln\left[q_i y_i\right] di\right),$$

where y_i and q_i denote the quantity and quality of product i. Output is consumed entirely such that $Y = C$. Expenditure minimization leads to the standard demand function

$$y_i = \frac{YP}{p_i}. \tag{2}$$

P is defined as the aggregate price index, which I normalize to 1.

4.2 Production

Firms, indexed by f, produce in product market i with a linear technology

$$y_{ift} = \varphi_f l_{ift},$$

where y_{ift} denotes output, l_{ift} labor hired, and φ_f (time-invariant) firm productivity. Importantly, φ_f differs across firms, which captures the notion that some firms are systematically more efficient at producing than others in all of their product lines, e.g., due to a better business plan. As in Aghion et al. (2023), I assume two productivity types, i.e., $\varphi_f \in \{\varphi^h, \varphi^l\}$ where $\varphi^h/\varphi^l > 1$, which I refer to as high- and low-type firms.

4.3 Static allocation

Taking the joint distribution of product qualities and firm productivity as exogenous in this section, I characterize the static allocations at the product, firm and aggregate levels.

4.3.1 Product level

Firms in product market i compete in prices (Bertrand competition). In equilibrium, only the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$ produces product i (henceforth, incumbent). Under Bertrand competition, the incumbent firm engages in limit pricing and sets its price equal to the quality-adjusted marginal costs of the follower (the firm with the second highest quality-adjusted productivity)

$$p_{if} = \frac{q_{if}}{q_{if'}} \frac{w}{\varphi_{f'}}, \tag{3}$$

where f' indexes the follower in product market i. The price that the incumbent sets is increasing in the quality gap between the incumbent and the follower, as eq. (3) shows. Defining the product markup as the output price over marginal costs, it follows

$$\mu_{if} \equiv \frac{p_{if}}{w/\varphi_f} = \frac{q_{if}}{q_{if'}} \frac{\varphi_f}{\varphi_{f'}}. \tag{4}$$

The incumbent's markup for product i is increasing in its quality and productivity gap. The price setting of the incumbent gives rise to the following profits for product i

$$\pi_{if} = p_{if} y_{if} - w l_{if} = Y \left(1 - \frac{1}{\mu_{if}}\right),$$

with labor demand for product i

$$l_{if} = \frac{Y}{w} \mu_{if}^{-1}. \tag{5}$$

Employment in product line i is decreasing in the markup.

4.3.2 Firm level

Firm employment is the sum of employment across the firm's product lines

$$l_f = \sum_{i \in N_f} l_{if} = \frac{Y}{w} \left(\sum_{i \in N_f} \mu_{if}^{-1} \right),$$

where N_f denotes the set of product lines in which firm f is the incumbent producer. Firm employment decreases in the markups within each product line but increases in the number of product lines. Hence, holding product markups constant, firms that produce in more product lines feature higher employment. Vice versa, holding the number of product lines constant, firms with higher product markups employ less labor. As sales are equalized across product lines, firm sales are given by $\left|N_f\right|Y \equiv n_f Y$, where n_f denotes the number of products firm f is producing. Hence, firms that produce in more product lines feature higher sales. Lastly, I define the firm markup μ_f as total firm sales over the wage bill wl_f.

4.3.3 Aggregate level

Integrating employment across firms or products yields the total workforce in production:

$$L_P = \int_f l_f df = \frac{Y}{w} \int_0^1 \mu_{if}^{-1} di. \tag{6}$$

Taking logs and integrating eq. (4), one obtains an expression for the wage

$$w = \exp\left(\int_0^1 \ln q_{if} di\right) \times \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right) \times \exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right). \tag{7}$$

To find an expression for aggregate output, insert eq. (7) into eq. (6) to obtain

$$Y = Q\Phi\mathcal{M}L_P, \tag{8}$$

where

$$Q = \exp\left(\int_0^1 \ln q_i di\right), \quad \Phi = \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right), \quad \mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_i^{-1} di\right)}{\int_0^1 \mu_i^{-1} di}.$$

Aggregate output Y depends on geometric averages of quality Q and productivity Φ as well as on misallocation $\mathcal{M}$ and production labor L_P. Misallocation arises from markup dispersion ($\mathcal{M}$ is bounded by unity from above) that is due to quality and productivity heterogeneity. The product of Q, Φ and $\mathcal{M}$ captures aggregate Total Factor Productivity (TFP).

4.4 Dynamic firm problem

Firms compete for product markets through innovation (R&D). Via expansion R&D, a firm improves the quality of a randomly selected product operated by a competitor, thereby becoming the highest-quality producer in a product line that is new to the firm. Following Peters (2020), firms also undertake internal R&D, which enables them to further raise the quality of products they already control after successful expansion. Internal R&D increases product markups according to eq. (4), creating a wedge between sales and wage-bill growth. This mechanism allows the model to match the distinct size-age relationship for sales (value added) and the wage bill—and their evolution over time—shown in Figure 1.

Item quality is improved step-wise such that every innovation (external or internal) increases q_i by a factor of λ.[9] Denoting by λ^{Δ_i} the relative qualities of incumbent and second-best firms within a product line, i.e.,

$$\lambda^{\Delta_i} = \frac{q_{if}}{q_{if\prime}}$$

and by $[\mu_i]$ a firm's set of markups across its product lines, firm profits follow

$$\pi_f(n, [\mu_i]) \equiv \sum_{k=1}^{n} \pi_i(\mu_k) = \sum_{k=1}^{n} Y \left(1 - \frac{1}{\mu_k}\right) = \sum_{k=1}^{n} Y \left(1 - \frac{1}{\lambda^{\Delta_k}} \frac{1}{\frac{\varphi_{f(k)}}{\varphi_{f'(k)}}}\right).$$

Incumbent firms choose the rate of expansion R&D, x_{it}, and the rate of internal R&D, I_{it}, for each of their product lines, i. When choosing x_{it} and I_{it}, firms take aggregate output Y_t, the real wage w_t, the share of lines operated by high-productivity firms S_t, the interest rate r_t and the rate of creative destruction τ_t as given. Note that S_t is predetermined in t and its evolution is characterized shortly. Denoting the time derivative by $\dot{V}_t^h()$, the value function of a high-productivity type firm (indexed by h) satisfies the following HJB equation:

$$\begin{aligned}
&r_t V_t^h(n, [\mu_i], S_t) - \dot{V}_t^h(n, [\mu_i], S_t) = \\
&\quad \sum_{k=1}^{n} \underbrace{\pi_i(\mu_k)}_{\text{Flow profits}} + \sum_{k=1}^{n} \tau_t \underbrace{\Big[V_t^h\left(n-1, [\mu_i]_{i\neq k}, S_t\right) - V_t^h(n, [\mu_i], S_t)\Big]}_{\text{Creative destruction}} \\
&\quad + \max_{[I_{it}, x_{it}]} \Bigg\{ \sum_{k=1}^{n} I_{kt} \underbrace{\Big[V_t^h\left(n, [[\mu_i]_{i\neq k}, \mu_k \cdot \lambda], S_t\right) - V_t^h(n, [\mu_i], S_t)\Big]}_{\text{Internal R\&D}} \\
&\quad + \sum_{k=1}^{n} x_{kt} \underbrace{\left[S_t V_t^h\left(n+1, [[\mu_i], \lambda], S_t\right) + (1-S_t) V_t^h\left(n+1, \left[[\mu_i], \lambda \cdot \frac{\varphi^h}{\varphi^l}\right], S_t\right) - V_t^h(n, [\mu_i], S_t)\right]}_{\text{Expansion R\&D}} \\
&\quad - \underbrace{w_t \Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right)}_{\text{R\&D costs}} \Bigg\}.
\end{aligned}$$

The value of a firm consists of flow profits, research costs, and three parts related to internal R&D, expansion R&D, and creative destruction. At the rate of creative destruction τ_t (determined in equilibrium), the firm loses one of its n products, in which case, it remains with $n-1$ products. At the optimally chosen rate I_{it}, internal R&D turns out successful (third row), and the firm charges a λ times higher markup on its product according to eq.

[9] As Aghion et al. (2023), I assume that the step size of quality improvements exceeds the productivity differential, $\lambda > \varphi^h/\varphi^l$. This assumption ensures that the firm with the highest quality version in a product line is the incumbent producer. Relaxing this assumption would give room for a race for incumbency between low-productivity entrants facing a high-productivity incumbent from which I abstract.

(4). Alternatively, at the optimally chosen rate x_{it}, expansion R&D is successful (fourth row), and the firm acquires a new product (n increases by one).

Productivity-type heterogeneity introduces novel elements to the value function of the firm. The share of product lines operated by each productivity type S_t is a state variable that firms keep track of to build markup expectations. When taking over a new product line through expansion R&D (fourth row), the probability of replacing a high-type incumbent is S_t, in which case the high-type entrant charges a markup of λ. With probability $1 - S_t$, the replaced incumbent is of the low type, and the high-type entrant charges a markup of $\lambda \cdot \varphi^h / \varphi^l$. Firms take S_t as given; however, they affect it through their expansion R&D efforts x_{it} in equilibrium. The HJB equation for a low-productivity firm follows the same structure and is listed in the Appendix, Section C.1. The term related to expansion R&D (fourth row) is distinct since markup expectations differ for low-productivity firms.

$\Gamma([x_{it}, I_{it}]; n, [\mu_i])$ denote the R&D costs. For their R&D activities, firms pay a cost of

$$\Gamma([x_{it}, I_{it}]; n, [\mu_i]) = \sum_{k=1}^{n} c(x_{kt}, I_{kt}; \mu_k) = \sum_{k=1}^{n} \left[\mu_k^{-1} \frac{1}{\psi_I} (I_{kt})^\zeta + \frac{1}{\psi_x} (x_{kt})^\zeta \right]$$

in terms of labor. $\zeta > 1$ ensures convexity of the cost function. R&D costs are additively separable to render a closed-form solution of the value function along the balanced growth path. ψ_I and ψ_x scale internal and external R&D costs and capture the R&D efficiency.[10]

The rate of creative destruction τ_t that firms take as given in their optimization problem is, in equilibrium, equal to the sum of expansion R&D rates and the rate of entry z_t

$$\tau_t = \int_0^1 x_{it} di + z_t. \tag{9}$$

Firm entry is determined as follows. Potential entrants produce a flow rate of entry z_t using a technology that is linear in labor: $z_t = \psi_z L_{Et}$, where ψ_z denotes the entry efficiency and L_{Et} research labor of entrants. Entrants improve the quality of a randomly selected product line. The productivity type is realized after entry and assigned with the exogenous probabilities p^h and $1 - p^h$, respectively. Entrants start with a one-step quality gap. When $z_t > 0$, the free entry condition requires that the expected value of firm entry equals the entry costs

$$p^h E[V_t^h(1, \mu_i)] + (1 - p^h) E[V_t^l(1, \mu_i)] = \frac{1}{\psi_z} w_t, \tag{10}$$

[10] Since product-line profits are concave in the markup, internal R&D incentives decline with successful innovations. This would imply slowing markup growth at older ages and faster wage-bill growth relative to sales (value added)—patterns not supported in Figure 1. I therefore follow Peters (2020) and scale internal R&D costs by the inverse markup, which removes the dependence of internal R&D on product-line markups while preserving its sensitivity to all other equilibrium forces.

where the expected value of entering as a high- or low-type firm is

$$\begin{aligned} E[V_t^h(1,\mu_i)] =& S_t V_t^h(1,\lambda) + (1-S_t)V_t^h(1,\lambda\times\varphi^h/\varphi^l) \\ E[V_t^l(1,\mu_i)] =& S_t V_t^l(1,\lambda\times\varphi^l/\varphi^h) + (1-S_t)V_t^l(1,\lambda). \end{aligned}$$

Firms' expansion and internal R&D efforts generate a two-dimensional distribution of quality and productivity gaps across product lines, denoted by $\nu_t(\Delta, \frac{\varphi_f}{\varphi_{f'}})$. These gaps determine product markups, so the distribution ν_t characterizes the overall markup distribution in the economy. I derive ν_t as a function of firm policies in Appendix C.2. Importantly, the marginal distributions of ν_t determine S_t, the share of product lines operated by high-productivity firms, which firms take as given in their optimization problem. The law-of-motion for S_t follows from ν_t:

$$\dot{S}_t = \sum_{i=1}^{\infty}\left[\dot{\nu}_t\left(i, \frac{\varphi^h}{\varphi^h}\right) + \dot{\nu}_t\left(i, \frac{\varphi^h}{\varphi^l}\right)\right] = S_t x_t^h + z_t p^h - S_t\tau_t. \tag{11}$$

Hence, S_t increases through the expansion of high-type incumbents into new product lines and the entry of new high-type firms, and decreases due to creative destruction.

Lastly, labor market clearing requires that production labor L_{Pt} and total research labor L_{Rt} add up to one, the aggregate labor endowment

$$1 = L_{Pt} + L_{Rt} = \frac{Y_t}{w_t}\int_0^1 \mu_i^{-1} di + \int_0^1 \left(\mu_i^{-1}\frac{I_{it}^{\zeta}}{\psi_I} + \frac{x_{it}^{\zeta}}{\psi_x}\right) di + \frac{z_t}{\psi_z}. \tag{12}$$

4.5 Balanced growth path characterization

I define a balanced growth path of the economy as follows.

Definition 1. *A balanced growth path (BGP) is a set of allocations $[x_{it}, I_{it}, \ell_{it}, z_t, S_t, y_{it}, C_t]_{it}$ and prices $[r_t, w_t, p_{it}]_{it}$ such that firms choose $[x_{it}, I_{it}, p_{it}]$ optimally, the representative household maximizes utility choosing $[C_t, y_{it}]_{it}$, the growth rate of aggregate variables is constant, the free-entry condition holds, all markets clear and the distribution of quality and productivity gaps ν_t is stationary.*

Along the balanced growth path, the economy can be characterized in closed form.

Proposition 1. *In the above setup, along a balanced growth path:*

1. *The value of a product line i for a firm of productivity type $d \in \{h, l\}$ is given by*

$$V_i^d(1,\mu_i,S) = \frac{1}{\rho+\tau}\left[Y_t\left(1-\frac{1}{\mu_i}\right) + \frac{\zeta-1}{\psi_x}x_i^{\zeta}w_t + \frac{\zeta-1}{\psi_I}I_i^{\zeta}w_t\mu_i^{-1}\right] \tag{13}$$

The value of a firm is the sum of its product line values $V^d(n, [\mu_i], S) = \sum_{k=1}^{n} V_i^d(1, \mu_k, S)$.

2. *Expansion R&D rates are firm productivity-type specific, i.e.,* $x_i \in \{x^h, x^l\}$. *Further, high-productivity firms expand faster than low-productivity ones* $x^h > x^l$.
3. *The share of product lines operated by high-productivity type firms* S *is determined by*

$$S\tau = Sx^h + zp^h. \tag{14}$$

4. *The growth rate of aggregate variables is given by*

$$g = \frac{\dot{Q}_t}{Q_t} = \Bigg(\underbrace{I}_{\textit{Incumbent internal R\&D}} + \underbrace{Sx^h + (1-S)x^l}_{\textit{Incumbent expansion R\&D}} + \underbrace{z}_{\textit{Entry}} \Bigg) \times \ln(\lambda). \tag{15}$$

Proof. The Appendix, Sections C.1, C.2 and C.3 contain the proofs. □

The value of a product line in eq. (13) consists of three terms: profits for a given markup and the continuation values of expansion and internal R&D. The sum of the three terms is discounted by the discount rate and the rate of creative destruction. The more impatient the household or the higher the risk of replacement, the lower the value of a product line. Importantly, expansion R&D rates are productivity-type specific. More productive incumbents charge higher markups, which increases the expected value of a product line. The optimality condition for expansion R&D (Appendix, eq. A-6) equates the expected value of a product line with the marginal cost of expansion R&D. Hence, in equilibrium, more productive firms pay a higher marginal cost of expansion R&D and choose $x^h > x^l$. By contrast, internal R&D rates are type-invariant, $I_i \equiv I$. Because expansion rates differ by type, the values of a product line and a firm are productivity-type specific as well.

Proposition 1 further shows that the share of product lines operated by high-productivity type firms is constant along the balanced growth path. The expression in eq. (14) directly follows by setting the law-of-motion for S_t in eq. (11) equal to zero. Eq. (14) captures that, along the balanced growth path, the share of product lines by high-productivity firms that fall victim to creative destruction is exactly equal to the share of product lines that high-productivity firms gain by entering new product lines through incumbent creative destruction or firm entry. Eq. (14) can further be rearranged to

$$S = \frac{zp^h}{(1-S)(x^l - x^h) + z}, \tag{16}$$

which shows that S depends on the difference in the expansion R&D rates between firm types, $x^l - x^h$. Holding firm entry z fixed, an increase in the expansion rate of high-productivity incumbents must be matched by an equal rise in the expansion rate of less-productive firms for S to remain constant. Note that positive firm entry is necessary for both firm types to co-exist in equilibrium where $x^l \neq x^h$ and S is constant at its interior solution.

Long-run growth results from R&D at the product level. This occurs through successful internal R&D or (incumbents' and entrants') creative destruction. The aggregate arrival rate of innovation is, hence, equal to the sum of the rates of creative destruction τ and internal R&D I. Multiplying the arrival rate by the log step size of innovation delivers the aggregate growth rate g, as shown in eq. (15) of Proposition 1. Since expansion R&D rates are heterogeneous, i.e., $x^h > x^l$, changes in the share of product lines operated by each productivity type, S and $1-S$, affect the aggregate growth rate. Along the balanced growth path, both τ and g are constant. I further characterize the aggregate labor income share, the TFP misallocation measure $\mathcal{M}$, and the aggregate markup analytically in the Appendix, Section C.2.

To find the balanced growth path, I jointly solve the optimality conditions of the firm (derived in Appendix C.1), the free entry condition, eq. (10), the labor market clearing condition, eq. (12), and the system of differential equations characterizing the distribution of productivity and quality gaps captured in eqs. (A-8) and (A-9). Appendix C.4 contains the details.

4.6 Firm dynamics

Firms lose products according to the same stochastic process as in Klette and Kortum (2004). However, in this model, firms add products at systematically different rates as optimally chosen expansion R&D rates vary with the firm's productivity type.[11] The following section derives productivity-type specific firm-size dynamics and survival probabilities.

4.6.1 Sales-age relationship

Firm sales are proportional to the number of products a firm produces. As such, successful expansion R&D increases firm sales. Since optimal expansion R&D rates are productivity-type specific, so is the sales-age relationship. For productivity type φ_f, the average log sales of surviving firms at age a_f relative to the average log sales of entrants is

$$E\left[\ln n_f Y | a_f, \varphi_f\right] - E\left[\ln n_f Y | 0, \varphi_f\right] = \underbrace{g \times a_f}_{\text{Aggregate growth}} + \underbrace{E\left[\ln n_f | a_f, \varphi_f\right]}_{\text{Avg. product count}},$$

where n_f is a firm's number of products. The probability of producing n products at age a conditional on survival is $(1-\gamma^j(a))\,(\gamma^j(a))^{n-1}$, where $\gamma^j(a) = x^j \frac{1-e^{-(\tau-x^j)a}}{\tau - x^j e^{-(\tau-x^j)a}}$ and $j \in \{h, l\}$.

[11]Therefore, the properties related to firm-size growth and survival in Klette and Kortum (2004) hold conditional on the firm type. In particular, conditional on the type, firm size, and growth are unrelated, as in Lentz and Mortensen (2008). For the unconditional firm-size and growth correlation, two forces are at play. On the one hand, young (small) firms tend to grow quicker due to the survival bias. On the other hand, more productive firms (with faster growth rates) are more likely to end up large. In the estimated model, the majority of the firms are of the high-productivity type. Hence, size is unrelated to growth for most firms.

Therefore, the type-specific sales-age relationship can be reformulated as

$$E\left[\ln n_f Y|a_f,\varphi_f\right]-E\left[\ln n_f Y|0,\varphi_f\right]=\underbrace{g\times a_f}_{\text{Aggregate growth}}+\underbrace{\left(1-\gamma^f(a_f)\right)\sum_{n=1}^{\infty}\ln n\times\left(\gamma^f(a_f)\right)^{n-1}}_{\text{Avg. product count}}. \tag{17}$$

4.6.2 Employment-age relationship

Average employment conditional on age and productivity type is equal to

$$E[\ln l_f|a_f,\varphi_f]=\ln\left(\frac{Y}{w}\right)+E\left[\ln n_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|a_f,\varphi_f\right].$$

Since $\frac{Y}{w}$ is constant, the type-specific employment-age relationship simplifies to

$$E[\ln l_f|a_f,\varphi_f]-E[\ln l_f|0,\varphi_f]=\underbrace{E\left[\ln n_f|a_f,\varphi_f\right]}_{\text{Avg. product count}}-\underbrace{\left(E\left[\ln\mu_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|0,\varphi_f\right]\right)}_{\text{Markup-age relationship}}, \tag{18}$$

where $E\left[\ln n_f|a_f,\varphi_f\right]$ is defined in eq. (17) and the markup-age relationship, $E\left[\ln\mu_f|a_f,\varphi_f\right]-E\left[\ln\mu_f|0,\varphi_f\right]$, is derived in Appendix C.6. As wages and output grow at the same rate along the balanced growth path, eqs. (17) and (18) further imply that the type-specific wage bill-age relationship is equal to the sales-age relationship less the markup-age relationship.

4.6.3 Firm survival

Firms that lose their last product exit the economy. Since firm expansion is type-dependent, so is firm survival. The survival function in Klette and Kortum (2004) holds conditional on the firm type, i.e., the share of high and low type firms surviving until age a_f is

$$\chi^h(a_f)=1-\tau\frac{1-e^{-(\tau-x^h)a_f}}{\tau-x^h e^{-(\tau-x^h)a_f}}\quad\text{and}\quad\chi^l(a_f)=1-\tau\frac{1-e^{-(\tau-x^l)a_f}}{\tau-x^l e^{-(\tau-x^l)a_f}}. \tag{19}$$

Hence, the share of high-type firms among surviving firms at age a_f is

$$s^h(a_f)=\frac{p^h\chi^h(a_f)}{p^h\chi^h(a_f)+(1-p^h)\chi^l(a_f)}, \tag{20}$$

which corresponds to the mass of high-type survivors relative to the total mass of survivors. Since $x^h>x^l$, it is easy to show that the sales-age relationship of high-productivity firms is steeper than the one of low-productivity ones and that the share of the former among surviving firms $s^h(a_f)$ increases in a_f.

4.6.4 Unconditional size-age relationship

The productivity-type specific size-age relationship and survival probabilities characterize the size-age relationship unconditional of the productivity type.

$$\begin{aligned} E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f\right] - E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0\right] = \qquad \qquad (21) \\ s^h(a_f)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^h\right] - E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right]\right)}_{\text{Size-age relationship (high-productivity type)}} \\ +\left(1-s^h(a_f)\right)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^\ell\right] - E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Size-age relationship (low-productivity type)}} \\ +\left(s^h(a_f)-s^h(0)\right)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right] - E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Correction term}}, \end{aligned}$$

The unconditional size-age relationship is equal to the type-specific size-age relationship, weighted by the share of each type among surviving firms at a given age, $s^h(a_f)$ and $1-s^h(a_f)$, plus a correction term. This last term corrects for size differences at entry between both productivity types. If size is measured using sales, the last term disappears since entrants of both productivity types start with equal sales. Hence, eq. (21) highlights that the unconditional size-age relationship steepens if the average size conditional on age a_f of either productivity type increases or the share of high-productivity firms among survivors, $s^h(a_f)$, rises since their average size conditional on age exceeds that of low-productivity firms.

4.6.5 Firm size distribution

The theory makes precise predictions about the firm-size distribution. I derive the firm-size distribution in Section C.7 in the Appendix. Denoting by M^h the mass of high-productivity type firms and by M the total mass of firms, the share of high-productivity type firms in the cross section is

$$S_{M^h} = \frac{M^h}{M}, \qquad (22)$$

and the firm entry rate equals

$$\text{Firm entry rate} = \frac{z}{M}. \qquad (23)$$

5 Balanced growth path application

This section applies the model to quantify the contributions of the different components of the dispersion decomposition to the rise in firm-size dispersion. To this extent, I estimate the model along two balanced growth paths. The initial balanced growth path captures firm-size dynamics and macroeconomic conditions during the 1990s. I then re-estimate model parameters to reflect the changes in firm-size dynamics and macroeconomic conditions with respect to the 2010s.

5.1 Initial balanced growth path

There are, in total, eight parameters in the model. The internal R&D efficiency ψ_I, the expansion R&D efficiency ψ_x, the innovation cost curvature ζ, the entry efficiency ψ_z, the step size of quality improvements λ, the productivity differential φ^h/φ^ℓ, the share of high-productivity type firms among entrants p^h, and the discount rate ρ. Three of these parameters are calibrated exogenously. First, the discount rate ρ is set to 0.05. Second, I follow Acemoglu et al. (2018) who set ζ equal to two based on evidence from the microeconometric innovation literature (Blundell et al., 2002; Hall and Ziedonis, 2001). Third, Figure 3 shows that firm size at entry has remained very stable over time, indicating no systematic changes in the nature of entrants. Therefore, I fix the share of high-productivity firms among entrants p^h at some exogenous level. Since the interquartile range as the measure of firm-size dispersion captures dispersion around the median firm, I set p^h to 0.5. Hence, one can interpret high- and low-productivity firms as above or below median firms at entry. The results in this paper are similar for alternative values of p^h.

The other five parameters are estimated, targeting the firm size-age relationship (for value added and the wage bill), cross-sectional firm-size dispersion, the firm entry rate, and aggregate TFP growth. Even though all parameters are identified jointly, there is a tight mapping between parameters and targets. This mapping is intuitively explained in the following and quantitatively explored in the sensitivity analysis after the estimation.

Matching the firm size-age relationship for value added and the wage bill disciplines the firms' R&D efficiencies ψ_x and ψ_I. In the model, successful expansion R&D raises firms' sales. In the estimation, ψ_x therefore adjusts expansion R&D costs to match a given target for the sales-age relationship. Since labor is the only production input in the model, firm sales equal value added, and I use the latter as the measure in the data. Moreover, the internal R&D costs govern firms' markup growth. Since the firm markup is the wedge between value added (sales) and the wage bill in the model, targeting the value added-age and the wage bill-age relationship jointly disciplines the markup-age relationship and, hence, the internal R&D efficiency ψ_I. The advantage of targeting the wage bill instead of markups is that the former is directly observed in the data. I target the average size of firms aged eight relative to the average size of entrants for the cohorts 1997 to 2001. Figure 1 quantified this number at 0.492 and 0.468 log points for value added and the wage bill, respectively. Eight years is long enough to reflect firm size growth and still allow for estimating separate balanced growth paths (one for the early cohorts and one for the latest cohorts) over the data coverage period from 1997 to 2017. The model matches the firm-size dynamics over the entire life cycle rather well, so the specific age targeted is not consequential.

To pin down the entry efficiency, ψ_z, and the productivity differential, φ^h/φ^ℓ, I target the en-

Table 2: Initial balanced growth path. Moments and parameters

	Data	Model
Moments		
Size-age relationship (log value added)	0.492	0.492
Size-age relationship (log wage bill)	0.468	0.468
Entry rate in %	14.8	14.8
Interquartile range of log value added	log(2)	log(2)
TFP growth g in %	1.5	1.5
Parameters		
ψ_x *Expansion R&D efficiency*		0.892
ψ_I *Internal R&D efficiency*		0.925
ψ_z *Entry R&D efficiency*		3.790
φ^h/φ^ℓ *Productivity gap*		1.019
λ *Step size of quality improvements*		1.051
Set exogenously		
ρ *Discount rate*		0.05
ζ *R&D cost curvature*		2
p^h *Share of high-type firms among entrants*		0.5

Notes: the size-age relationship measures the average size (value added or wage bill) of surviving firms at age eight relative to the average size of entrants in logs. The targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

try rate and total cross-sectional firm-size dispersion. ψ_z governs the labor cost of a given flow rate of entry so that the entry rate identifies ψ_z. Because the entry rate shapes the firm-age distribution—the second component of the dispersion decomposition in eq. (1)—targeting total firm-size dispersion jointly with the entry rate and the size-age relationship implies that size dispersion conditional on age—corresponding to the third component of the decomposition—is disciplined residually. The productivity differential φ^h/φ^ℓ then adjusts to match this implied dispersion conditional on age, as a larger differential generates greater heterogeneity in expansion rates across productivity types.[12]

For the entry rate, I target its value in 1997, which stands at 14.8% as previously shown in Figure 5a. For the measure of total size dispersion, I target the interquartile range of log value added within two-digit industries in Figure 4b. Size dispersion in the data is enormous. To make the size dispersion in the data comparable to the model, I normalize a firm's value added by its respective value added at entry. This normalization equalizes firm size at entry in the data and resembles the fact that, as common in Klette and Kortum (2004)

[12] An alternative would be to target the entire firm-age distribution jointly with total firm-size dispersion and the firm size-age relationship. However, the age distribution is not summarized by a single statistic in the way the entry rate is. Targeting the entry rate also ensures that the balanced growth paths are quantitatively consistent with the observed macro trend of declining firm entry. In addition, since firm age in the data is truncated for old firms, I do not observe the full age distribution.

style models, all firms start with one product. Without this normalization, cross-sectional size heterogeneity in the data that is due to size dispersion at firm entry is loaded on the productivity differential φ^h/φ^ℓ. I take the average within-industry interquartile range over the most recent years (2010–2017) and subtract the observed increase in Figure 4b. This results in an interquartile range of 0.742, i.e., the 75th-percentile firm is roughly $e^{0.742} \approx 2.1$ times as large as the 25th-percentile firm in 1997. In the model, the interquartile range is restricted to integer values.[13] Since the value in the data is very close to log(2), I target this value for the initial balanced growth path.[14]

Lastly, to pin down the step size of quality improvements, λ, I target aggregate TFP growth. In the model, the aggregate growth rate g in eq. (15) directly depends on λ. I target an aggregate growth rate of 1.5%, the average over the period 1995–2005 in Figure 5b. All targets are summarized in Table 2.

The estimation follows a two-step approach. In the first (global) step, the algorithm computes the sum of squared percentage deviations from the targeted moments for a large Sobol sequence of parameter vectors. All targets receive equal weights. In the second (local) step, I take the best candidates from the first step and perform a local search. The local search, again, minimizes the distance from the targets. The estimation is robust, i.e., the best parameter vectors from the second step converge to the same parameter values.

Table 2 shows the estimation results. The model replicates all targeted moments exactly. Two parameters can be interpreted intuitively: successful innovation increases product quality by 5.1%, and the productivity differential φ^h/φ^ℓ equals roughly 2%. The productivity advantage is relatively small, which relates to the fact that I target firm-size dispersion in the data that is not accounted for by size dispersion at entry.[15] Nevertheless, firms of the two types differ significantly over the life cycle, as shown below.

Along the balanced growth path, high-productivity firms constitute 56% of all firms in the cross-section, S_{M^h} in eq. (22), and capture 65% of total sales (value added), as measured by S. These numbers are larger than their firm shares and sales shares among entrants ($p^h = 0.5$) due to high-productivity firms expanding faster into new product lines than less productive ones: $x^h - x^\ell = 0.053$. This is reflected in their size-age relationship. For the high-productivity type, average value added of surviving firms at age eight is 0.58 log points higher than the average among entrants, compared with only 0.39 log points for the low-type. Figure

[13]For example, if the firm at the 25th percentile of the cross-sectional value-added distribution produces one product, the interquartile range of log value added is equal to log(2) if the 75th percentile firm produces two products, log(3) if the 75th percentile firm produces three products and so forth.

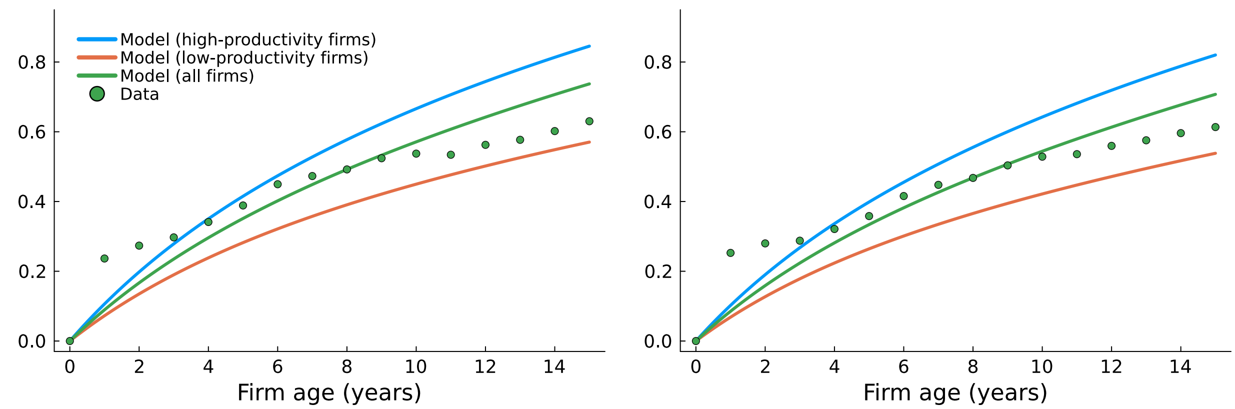
[14]To improve the computational efficiency of the estimation, I target a value for the standard deviation of log value added instead of the interquartile range and check afterward that the latter equals log(2) along the balanced growth path. The advantage of the former is that it provides finer gradients of the objective function in the parameter search. Computing the empirical counterpart of the standard deviation would require dropping firms with negative value added.

[15]Another reason why the productivity gap is relatively small compared to, e.g., Aghion et al. (2023) is that their model abstracts from firm markup growth, so the aggregate markup level, which is targeted in their estimation, disciplines the productivity differential.

Figure 6: Firm size-age relationship

(a) Value added

(b) Wage bill



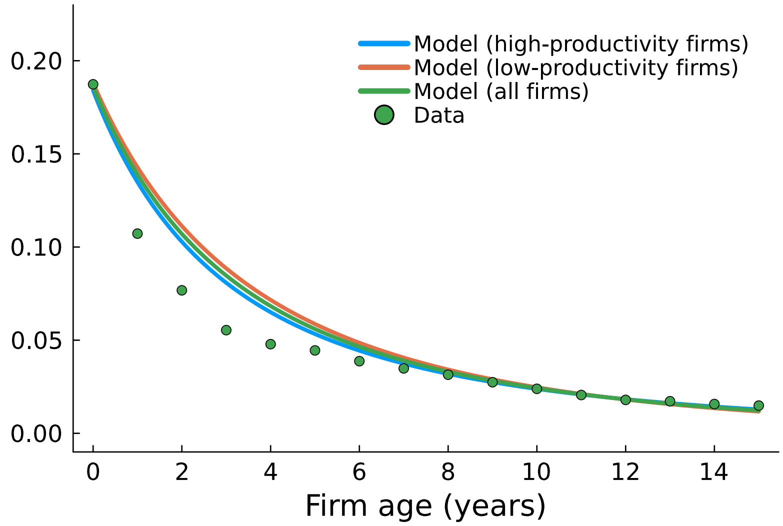
Notes: the figure shows the average value added (left) and wage bill (right) relative to the average among entrants (in logs) as a function of firm age in the model (initial balanced growth path) and the data (cohorts 1997–2001 in Swedish registry data). The relative size at age eight has been targeted.

6 shows the size-age relationship for value added and the wage bill for each productivity type over the entire life cycle. The figure clearly illustrates the heterogeneity in size-age profiles. For value added, the difference between both productivity types accumulates to roughly 0.3 log points after 15 years. Figure 6 includes the size-age relationship in the data (cohorts 1997–2001) and its respective model counterpart (the unconditional size-age relationship). The model slightly misses the concavity of the trajectories for the first two years; however, the fit over the entire life cycle is overall good despite being untargeted (except for age eight).

The model also performs well along other untargeted dimensions. Figure 7 shows the firm exit rates by age in the model and the data (cohorts 1997-2001). The model slightly overshoots the exit rates for the first few years, consistent with undershooting firm size growth in Figure 6 for these years. However, the overall fit over the entire life cycle is good. In the model, higher expansion rates by the more productive firms translate into lower exit rates, as Figure 7 shows. The differences in exit rates accumulate over firms' life cycles. The share of high-productivity firms that survive until age 15, characterized by $\chi^h(15)$ in eq. (19), is about five percentage points higher than for low-productivity ones.

Next, I conduct a set of comparative-statics exercises around the initial balanced growth path to illustrate how different model parameters affect cross-sectional firm-size dispersion. I vary each parameter individually by 20%, choosing the direction of adjustment such that firm-size dispersion increases relative to the initial balanced growth path. The left panel of Table 3 reports selected equilibrium outcomes. As my summary measure of firm-size dispersion, I use the standard deviation of log value added across firms, denoted σ_{VA}. The right panel of Table 3 decomposes the resulting change in σ_{VA} into its components following eq. (1). Each entry reports the share (in percent) of the total change in dispersion (relative to the initial balanced growth path) attributable to changes in the firm size-age relationship, $E[\ln s|a] - E[\ln s|0]$,

Figure 7: Exit rates by firm age



Notes: the figure shows the firm exit rates as a function of age in the estimated model (initial balanced growth path) and the data (cohorts 1997–2001). The exit rates are computed as the increments in a Kaplan-Meier survival function.

the age distribution, $p(a)$, or within-age firm-size dispersion, $\mathrm{Var}(\ln s|a)$.[16] Because the decomposition is not additive, the shares do not necessarily sum to 100%.

First, Table 3 shows that a reduction in expansion costs (an increase in expansion efficiency, ψ_x) raises the expansion rate of high-productivity firms from 0.144 to 0.177 and that of low-productivity firms from 0.091 to 0.105. The firm entry rate, z/M, falls from 14.8% to 13.4%, while the share of firms aged ten or older, $s_{a\geq 10}$, rises from 21.2% to 22.4%. These adjustments are associated with an increase in the standard deviation of log firm size from 0.50 to 0.63. The right panel of Table 3 indicates that the resulting rise in firm-size dispersion is driven by a steepening of the firm size-age relationship (37.8%) and an increase in size dispersion conditional on age (36.9%). Changes in the firm-age distribution account for a smaller share (21.2%). Taken together, the decomposition indicates that lower expansion costs increase firm-size dispersion by jointly steepening the size-age relationship and increasing size dispersion conditional on age.

Second, a rise in the productivity gap, φ^h/φ^ℓ, raises the expansion rate of high-productivity firms while lowering that of low-productivity firms. This divergence in expansion rates naturally amplifies size dispersion within age brackets: firms of the same age become more unequal because some expand faster and others more slowly. Accordingly, the largest share of the resulting increase in overall dispersion comes from changes in size dispersion conditional on age (39.4%), whereas movements in the size-age relationship contribute only modestly (23.5%).

Lastly, an increase in entry costs (a decline in entry efficiency, ψ_z) substantially reduces

[16]For the firm size-age relationship, to highlight the effects of the displayed changes in firms' expansion rates x^h and x^ℓ, I hold all other equilibrium variables fixed at their initial levels in this section.

Table 3: Comparative statics in the initial balanced growth path

	x^h	x^ℓ	$\frac{z}{M}$	$s_{a\geq 10}$	σ_{VA}	Decomposing changes in σ_{VA} (in %) $E[\ln s\|a] - E[\ln s\|0]$	$p(a)$	$\text{Var}(\ln s\|a)$
Initial BGP	0.144	0.091	0.148	0.212	0.5			
Parameters								
$\psi_x \uparrow$	0.177	0.105	0.134	0.224	0.63	37.8	21.2	36.9
$\varphi^h/\varphi^\ell \uparrow$	0.15	0.085	0.144	0.217	0.53	23.5	35.2	39.4
$\psi_z \downarrow$	0.182	0.1	0.075	0.295	1.18	12.4	58.8	18.6

Notes: the left table shows balanced growth path (BGP) outcomes (columns) for different model parametrizations (rows). x^h and x^ℓ denote the expansion R&D rates of high- and low-productivity firms, $\frac{z}{M}$ the entry rate, $s_{a\geq 10}$ the share of firms aged ten years or older defined by the cdf $p^h(1-\chi^h(a_f)) + (1-p^h)(1-\chi^\ell(a_f))$ and σ_{VA} the standard deviation of log value added across firms. The parameters ψ_x and ψ_z denote the expansion and entry R&D efficiency, respectively. φ^h/φ^ℓ is the productivity differential. The size of the parameter change is 20% (for φ^h/φ^ℓ the change refers to the parameter value minus one). The direction of the change is chosen to generate an increase in σ_{VA} relative to the initial BGP. The right table decomposes the changes in σ_{VA} according to eq. (1). Each number indicates the share (in percent) of the change in σ_{VA} that is due to changes in the size-age relationship, the age distribution and size dispersion conditional on age, respectively.

the firm entry rate to 7.5% and increases the share of firms aged ten or older to 29.5%. Consistent with this pattern, most of the resulting increase in size dispersion is accounted for by the shift in the firm-age distribution (58.8%).

5.2 New balanced growth path

To quantify the contributions of the different components of the dispersion decomposition to the observed rise in firm-size dispersion, this section estimates the model along a new balanced growth path that—relative to the previous one—replicates the observed changes. I target changes in the same five moments used in the initial estimation. The first two targets capture the steepening of the size-age relationship (value added and the wage bill) depicted in Figure 1. Specifically, I target the average for the 2002–2006 and 2007–2011 cohorts. At age eight, the average log value added of surviving firms exceeds the average among entrants by 0.638 log points; for the wage bill, the corresponding figure is 0.623 log points. In the initial balanced growth path, these differences were 0.492 and 0.468 log points, respectively. As the third moment, I target the fall in the firm entry rate from 15% to 11% documented in Section 3.3. Fourth, I match the total change in firm-size dispersion. Figure 4b shows that the interquartile range of value added increased by 0.74 log points. In the model, I match this by raising the interquartile range from log(2) in the initial balanced growth path to log(5) in the new one[17]—an increase of 0.92 log points, which slightly overshoots the observed value. I target log(5) instead of log(4) (which would undershoot the change) to give the firm-age distribution and size dispersion conditional on age more room to explain the increase in total size dispersion. Even with this conservative target, the steepening of

[17]As described for the initial balanced growth path, I target a value for the standard deviation of log value added in the estimation and check afterward that the generated interquartile range equals log(5).

the firm size-age relationship turns out to be the largest driver. Lastly, I target the decline in the aggregate growth rate from 1.5% to 1.1% shown in Figure 5b. I allow all five parameters of the previous estimation to change. In this setup, the steepening of the firm size-age relationship is matched directly, changes in the firm-age distribution are disciplined by the targeted decline in the entry rate, and the change in size dispersion conditional on age is determined residually through the targeted rise in total dispersion.

Table 4: New balanced growth path. Moments and parameters

	Data	Model	Initial BGP
Moments			
Size-age relationship (log value added)	0.638	0.638	0.492
Size-age relationship (log wage bill)	0.623	0.623	0.468
Entry rate in %	11	11	14.8
Interquartile range of log value added	log(5)	log(5)	log(2)
TFP growth g in %	1.1	1.1	1.5
Parameters			
ψ_x *Expansion R&D efficiency*		1.966	0.892
ψ_I *Internal R&D efficiency*		1.181	0.925
ψ_z *Entry R&D efficiency*		6.395	3.790
φ^h/φ^ℓ *Productivity gap*		1.021	1.019
λ *Step size of quality improvements*		1.036	1.051

Notes: the size-age relationship measures the average size (value added or wage bill) of surviving firms at age eight relative to the average size of entrants in logs. The targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

Table 4 captures all targeted moments and parameter estimates of the new balanced growth path. The estimation replicates the changes in all five moments well. The last column of the table compares the estimated parameters with their respective values in the initial balanced growth path. The identified parameter changes are consistent with the literature that studies the drivers behind recent macroeconomic trends. The estimation highlights a fall in the cost of expanding into new product lines (rise in the expansion R&D efficiency) and an increase in the productivity gap between high- and low-type firms. Cao et al. (2022) and Aghion et al. (2023) arrive at similar conclusions. Further, Table 4 points to innovations becoming more incremental as the step size of quality improvements declines. This mechanism is emphasized by Olmstead-Rumsey (2019).[18] The estimated increase in the entry R&D efficiency in Table 4 is somewhat surprising at first. The reason behind the increase in the entry efficiency is that higher expansion R&D rates, x^h and x^ℓ, induced by the fall in expansion costs reduce the equilibrium entry rate z needed to clear the labor market characterized by eq. (12).

[18]Note that the decline in research productivity documented by Bloom et al. (2020) does not necessarily contradict the estimated increase in expansion efficiency, since output per innovation—captured by the step size of quality improvements—declines across balanced growth paths. Therefore, the estimates point to the output rather than the input side of innovation as the driver of declining R&D productivity.

Quantitatively, this mechanism even lowers entry by more than the targeted decline. The estimation therefore compensates by reducing entry costs.[19]

To quantify the contributions of changes in the firm size-age relationship, the firm-age distribution, and size dispersion conditional on age to the overall increase in dispersion, I again apply eq. (1), analogous to the decomposition in Table 3. The results show that, across balanced growth paths, 34.4% of the rise in σ_{VA} is attributable to the steepening of the firm size-age relationship, 28.5% to changes in the age distribution, and 26.2% to an increase in size dispersion conditional on age. Therefore, the steepening of the firm size-age relationship accounts for the largest share of the rise in dispersion, however, firm aging and rising dispersion conditional on age also contribute significantly. Overall, the increase in firm-size dispersion reflects broadly balanced contributions across all three margins of firm dynamics.

6 Extensions

6.1 Transitional dynamics

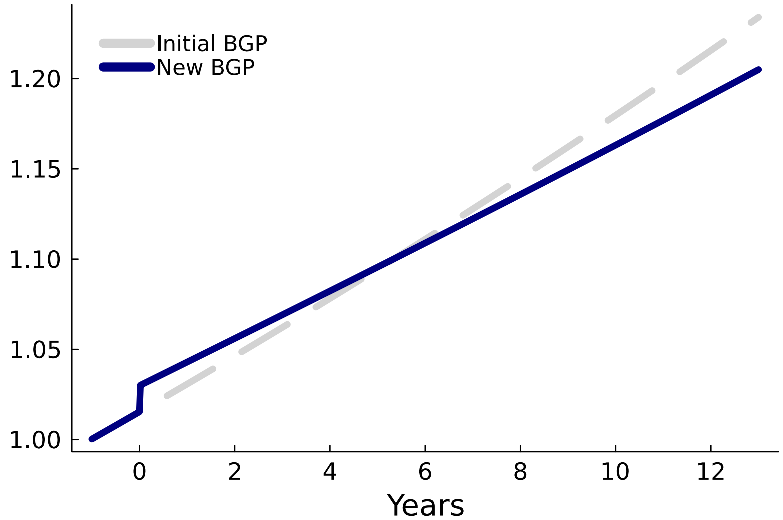
What are the welfare effects associated with the structural changes? To evaluate the changes in welfare, I solve for the full transition path to the new balanced growth path.

The algorithm to solve for the transition path works as follows. I solve for firms' policy and value functions from the new balanced growth path backward for a guessed sequence of wage growth, interest rates, and market shares of high-productivity firms, S_t. I then use the obtained policy functions over the transition period to simulate the two-dimensional distribution of quality and productivity gaps forward, starting from the initial balanced growth path. Using the evolution of this distribution over the transition, as well as market clearing and optimality conditions, I back out the implied sequences of wage growth, interest rates, and S_t. The transition path is the fixed point between the guessed and implied sequences. I outline the algorithm in detail in the Appendix, Section D.

Figure 8 shows the transitional dynamics for aggregate output (consumption) during the transition towards the new balanced growth path. Output along the initial balanced growth path is added for comparison in the dashed line. All parameter changes are introduced as one-time shocks in period zero. Output reacts positively on impact, resulting in a short-run boom in the economy relative to the initial balanced growth path. However, the growth burst is short-lived. Already after six years, the output path crosses with the trend of the initial balanced growth path and remains below. Figure 9 provides a closer look at equilibrium outcomes that drive the aggregate growth rate. As expansion costs fall, the expansion R&D rates of both high- and low-productivity firms increase in period zero. However, the increase

[19] A common interpretation of rising entry costs is that incumbents have erected barriers to entry. An alternative view is that improved access to finance has reduced entry costs for young firms. The model's estimation favors the latter interpretation.

Figure 8: Transitional dynamics for output

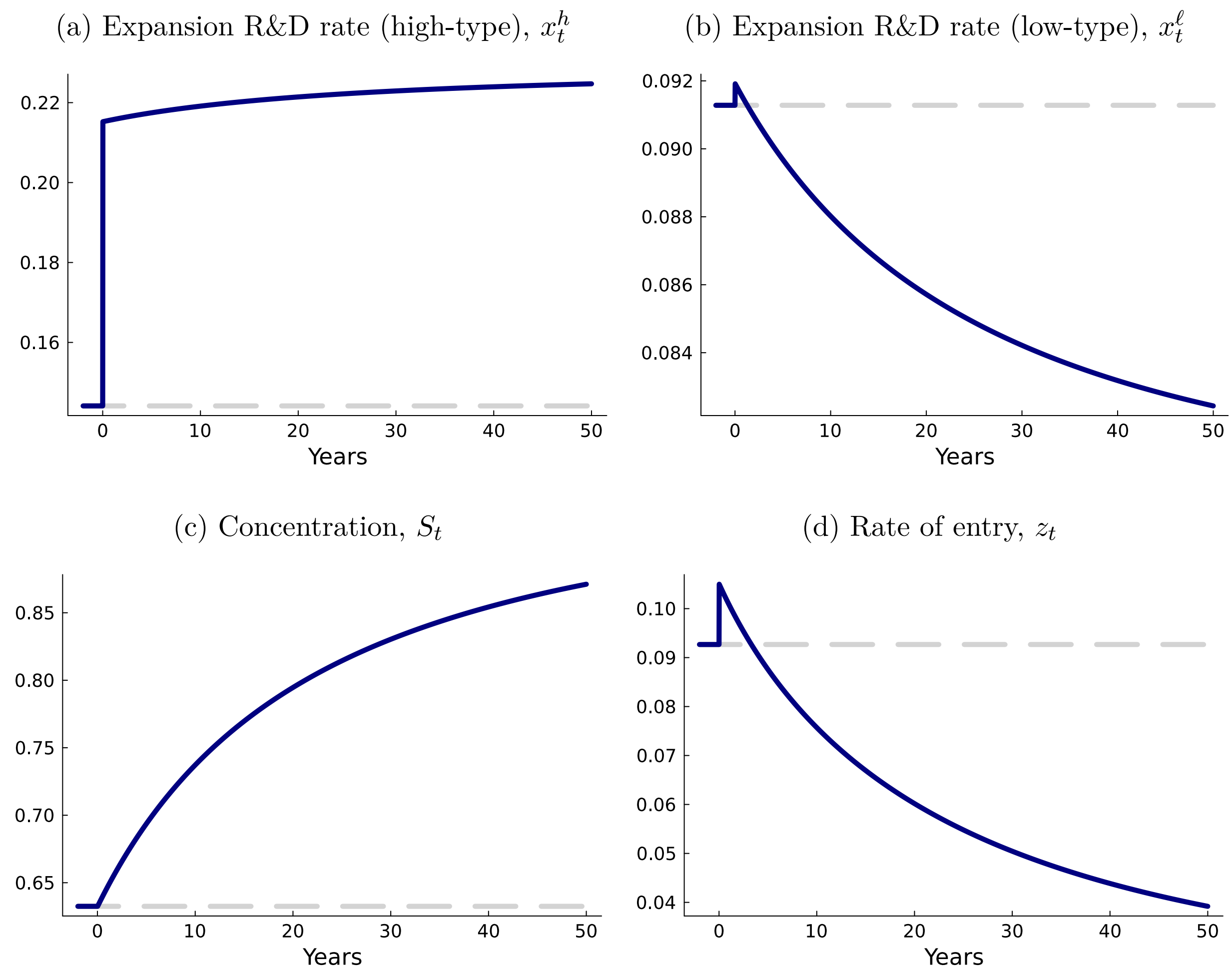


Notes: the figure shows output (consumption) along the transition towards the new balanced growth path. Parameters of the initial and new balanced growth path (BGP) are shown in Table 4. All parameter changes are introduced in period zero. Output is normalized to unity in year minus one for both BGPs.

is differently pronounced. Expansion rates of high-productivity firms increase substantially, whereas the increase is muted for low-productivity firms. Expansion rates continue to increase over time for high-productivity firms but gradually decrease for less productive ones. The increasing gap in expansion rates results in a reallocation of market shares towards the high-productivity firms: the share of product lines operated by the high type, S_t, gradually increases over time. The bottom right panel shows the flow rate of entry z_t. The increase in the entry R&D efficiency increases z_t on impact; however, the increase is of short duration, as the rise in the expansion rate x^h puts downward pressure on firm entry through labor market clearing.

That output increases in the short run relative to the initial balanced growth path motivates the question of what the net effect on welfare is. To quantify the change in welfare, I compute the permanent consumption change (in percent) along the initial balanced growth path that makes the representative consumer as well off as with the obtained consumption stream over the transition toward the new balanced growth path. I find that welfare decreases by 4.95%. Hence, the long-run fall in output growth dominates the short-run increase. If one were to compare welfare of the two balanced growth paths without taking the transition into account, the consumption equivalent change ξ is determined by $\ln(1+\xi) = (g^{\text{new}} - g^{\text{initial}})/\rho$, where g^{new} and g^{initial} refer to the long-run aggregate growth rate in each balanced growth path. Given that g declines by 0.4 percentage points across the balanced growth paths and ρ equals 0.05, the welfare loss amounts to 7.69% ($\xi = -0.0769$). Hence, the transition alleviates roughly one-third of the long-run welfare loss.

Figure 9: Transitional dynamics for equilibrium outcomes



Notes: the figure shows the response in equilibrium outcomes during the transition period towards the new balanced growth path. Parameters of the initial and new balanced growth path are shown in Table 4. All parameter changes are introduced in period zero. The dashed line indicates the initial balanced growth path.

6.2 Determinants of the firm size-age relationship

Is the steepening of the firm size-age relationship driven by an increase in average firm size conditional on age, holding the productivity composition of survivors fixed, or by changes in survivor composition? This section disentangles these two margins using the structural model.

The decomposition in eq. (21) highlighted that the unconditional firm size-age relationship equals the weighted average of the productivity type-specific size-age relationship, where the weights reflect the share of each type among survivors at a given age. The productivity-specific size-age relationship and survivor shares both depend on expansion R&D rates: higher expansion rates allow either type to enter more product markets, raising firm size conditional on age and increasing survival itself. The transition-path analysis showed that

Table 5: Determinants of the firm size-age relationship

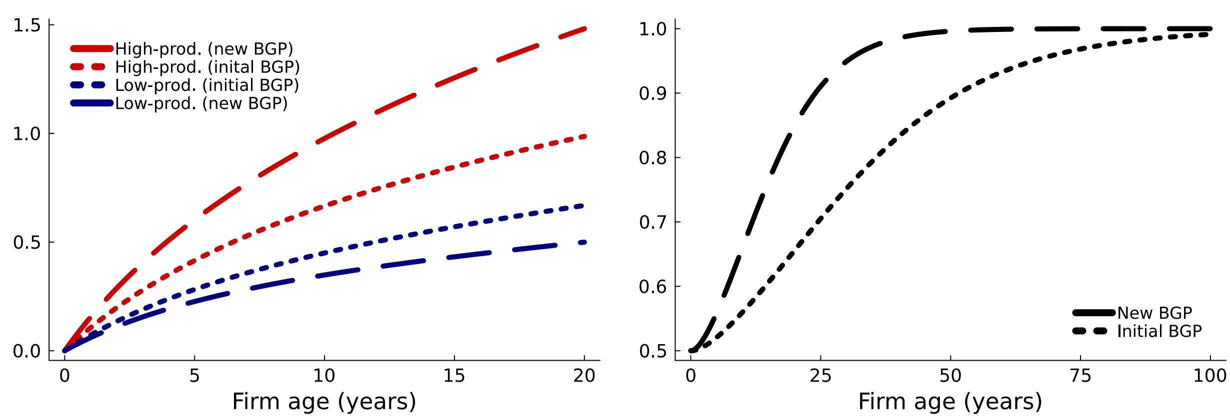
	Initial BGP	New BGP
Expansion R&D rate (high-prod.) x^h	0.144	0.228
Expansion R&D rate (low-prod.) x^ℓ	0.091	0.08
Share (high-prod.) among surviving firms s^h	0.543	0.619
Size-age relationship (high-prod.)	0.577	0.842
Size-age relationship (low-prod)	0.391	0.307

Notes: the table shows the expansion R&D rates of high- and low-productivity firms (first two rows), the share of high-productivity firms among firms alive at age eight (third row) and the average size (value added) of surviving firms at age eight relative to the average size of entrants for high- and low productivity firms in logs (last two rows). Parameter values of the initial and new balanced growth path (BGP) are shown in Table 4.

expansion rates for both types increase in the short run; over the longer run, the rate for high-productivity firms continues to rise, while that for low-productivity firms gradually declines. Table 5 reports for both balanced growth paths the productivity-type specific expansion R&D rates, size-age relationship and survivor shares. For high-productivity firms, the expansion rate rises from 0.144 to 0.228, while for low-productivity firms it falls from 0.091 to 0.08. As a result, the average size (value added) at age eight relative to the average size of entrants increases from 0.577 to 0.842 log points for high-productivity firms and decreases from 0.391 to 0.307 log points for low-productivity firms. At the same time, higher expansion rates boost survival for high-productivity firms, while the opposite holds for low-productivity firms. Consequently, the share of high-productivity firms among eight-year-old survivors, $s^h(8)$, rises from 0.543 to 0.619. Therefore, eq. (21) implies that the steepening of the unconditional size-age relationship is driven by two forces: more productive firms have become larger conditional on age and constitute a higher share of the survivor pool.

Figure 10: Determinants of the firm size-age relationship

(a) Size-age relationship (b) High-prod. type among surviving firms



Notes: the left figure shows the average firm size (value added) of surviving firms relative to the average size of entrants (in logs) for high and low productivity firms as a function of firm age. The right figure shows the share of high productivity firms among surviving firms as a function of firm age.

Figure 10 illustrates this mechanism for all firm ages. The left panel shows the firm size-age relationship (value added) for high- and low-productivity firms. At age eight, the values align with those reported in Table 5. The increase in the expansion R&D rates of high productivity firms increases their average size conditional on age, while the opposite holds true for low productivity firms. The right panel plots the share of high-productivity firms among surviving firms, s^h, as a function of firm age. This share starts at 0.5 at age zero, reflecting the share of high-type firms among entrants (p^h), and gradually converges to one. This convergence occurs because high-type firms, owing to their higher expansion R&D rates, are more likely to survive. In other words, although few firms survive to older ages, those that do are increasingly of the high-productivity type. As the gap in expansion R&D rates widens across balanced growth paths, this convergence accelerates, implying a higher share of high-productivity survivors at every firm age.

6.3 Alternative parameterizations of p^h

In the main analysis, the parameter p^h capturing the share of high-productivity firms among entrants was set exogenously to 0.5. This value was motivated by the fact that the interquartile range captures size dispersion around the median firm. I relax this assumption in this section. In particular, I estimate the initial and new balanced growth path for alternative parameterizations of p^h and, as before, quantify the contributions of the firm size-age relationship, the firm age distribution and size dispersion conditional on firm age to the total changes in cross-sectional firm-size dispersion.

Table 6 shows the estimated parameters in the initial and new balanced growth path for two alternative parameterizations of p^h, in particular $p^h = 0.4$ and $p^h = 0.6$. The results are similar to before. In particular, both alternative estimations ask for a significant increase in the expansion R&D efficiency. One notable observation is that the estimation asks for a larger productivity gap in the initial balanced growth path when $p^h = 0.6$ than when $p^h = 0.4$. In other words, the model implies a greater productivity differential between the top 60% and bottom 40% of entrants than between the top 40% and bottom 60%. Likewise, the increase in the productivity gap across balanced growth paths is more pronounced when p^h is set to 0.6. For p^h set to 0.4, the estimation even asks for a slight decrease in the productivity gap. Therefore, the estimation suggests that productivity dispersion increased primarily between the top 60% and bottom 40% of entrants, rather than between the top 40% and bottom 60%. This finding is notable, as the existing literature has instead emphasized the role of top firms in accounting for recent macroeconomic trends.

I repeat the decomposition of the rise in cross-sectional size dispersion for the alternative values of p^h. For $p^h = 0.4$, changes in the size-age relationship, the firm-age distribution, and dispersion conditional on age account for 31.8%, 29.8%, and 25.2% of the increase, respectively. For $p^h = 0.6$, the corresponding contributions are 35.8%, 28.2%, and 26.2%. Hence, the main conclusion remains unchanged: the steepening of the firm size-age relationship accounts for the largest share of the increase in dispersion, while all three margins

Table 6: Estimation for alternative values of p^h

	Initial BGP		New BGP	
	Data	Model	Data	Model
$\boldsymbol{p^h = 0.4}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.637
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.623
Entry rate in %	14.8	14.8	11	11
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.894		2.065
ψ_I *Internal R&D efficiency*		0.936		1.357
ψ_z *Entry R&D efficiency*		3.786		6.329
φ^h/φ^ℓ *Productivity gap*		1.017		1.014
λ *Step size of quality improvements*		1.051		1.035
$\boldsymbol{p^h = 0.6}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.649
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.632
Entry rate in %	14.8	14.8	11	11
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.888		1.814
ψ_I *Internal R&D efficiency*		0.905		0.913
ψ_z *Entry R&D efficiency*		3.792		6.522
φ^h/φ^ℓ *Productivity gap*		1.022		1.038
λ *Step size of quality improvements*		1.051		1.038

contribute significantly. Note that in the estimation, the size-age relationship is targeted only at age eight. As a result, the exact magnitudes in the decomposition can vary slightly across estimations.

6.4 Discussion on markups

Evidence on markups for the Swedish economy is scarce. Sandström (2020) provides an estimate of the aggregate markup using, as I do, administrative data on the balance sheets of Swedish firms. The author reports an estimate of about 15% for the sales-weighted aggregate markup and 5% for the unweighted one. These estimates are lower than common markup estimates for the U.S. (De Loecker et al., 2020); however, a direct comparison is difficult due to differences in, e.g., the data coverage or the institutional setting. Importantly, the markup estimates in Sandström (2020) are stable, displaying no systematic trend over time. This stability motivates the exclusion of the aggregate markup from the main analysis.[20]

7 Conclusion

This paper provides a decomposition of the rise in cross-sectional firm-size dispersion into distinct, observable margins of firm dynamics. Size dispersion can increase because the average size of surviving firms rises more steeply with age, because the distribution of firms shifts toward older firms, or because size heterogeneity conditional on age expands. The same increase in size dispersion can therefore arise from different underlying margins of firm dynamics, making it essential to distinguish between them. For Sweden, I show that all three margins have contributed to the overall rise in firm-size dispersion. Importantly, rising size dispersion conditional on firm age—despite receiving relatively little attention in the literature—accounts for a share of the increase comparable to that of changes in the age distribution.

More broadly, applying such a decomposition to the rise in firm-size dispersion in other countries would be a natural next step. For the United States, Hopenhayn et al. (2022) and Karahan et al. (2024) highlight a shift in the firm-age distribution as the driver behind recent macroeconomic trends. A systematic analysis of size dispersion conditional on firm age would help determine how much of the increase in cross-sectional firm-size dispersion reflects changes in the age distribution versus greater dispersion within age groups.

In the model, changes in expansion costs, entry costs, and ex-ante productivity dispersion map into the firm size-age relationship, the firm-age distribution, and size dispersion conditional on age. The primary objective is to quantify the contribution of these endogenous margins, rather than to uniquely identify the exogenous forces driving them. Nevertheless, the welfare implications of changes in these margins depend on the underlying exogenous forces that generate them. For example, a steepening of the size-age relationship driven by rising fixed operating costs—rather than declining expansion costs—would operate primarily through stronger selection and would not generate offsetting growth effects. In that case, the welfare consequences of rising firm-size dispersion could be substantially more negative than those estimated here.

[20] In the model, the (cost-weighted) markup in the estimated initial balanced growth path stands at 7.1%, well within the estimates provided in Sandström (2020) and even declines slightly in the new balanced growth path. Despite high-productivity firms with relatively high markups capturing additional market shares in the new balanced growth path, the fall in the step size of quality improvements puts downward pressure on within-firm markups.

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Online Appendix for

"Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends"

[FOR ONLINE PUBLICATION ONLY]

A Data

The main data set, *Företagens Ekonomi* (FEK), covers information from balance sheets and profit and loss statements for the universe of Swedish firms. From this data, I obtain the main variables of interest, namely value added (*Förädlingsvarde*, variable name: *Foradlingsvarde*) and the wage bill (*Lönekostnader*, variable name: *KonstnaderForLonerOchAndraErsattn*). In the FEK code-book by Statistics Sweden (SCB), these variables are defined as follows.[21] SCB defines value added as a firm's production value minus costs for purchased goods and services used as inputs in production (salaries and social security contributions not included). SCB's measure of value added captures the firm's contribution to the gross domestic product. The measure of the firm's wage bill includes salaries and other remuneration, including severance pay. As described in the main text, I focus on firms in the private sector. These firms have a legal type (variable name: *JurForm*) less than 50 or equal to 96.

To measure firm-size dispersion within industries, I use the industry classification (SNI codes) provided by SCB. The industry classification changed twice between 1997 and 2017, once in 2002 and once in 2007. The changes in industry classification mainly affect the more detailed codes at the five-digit level, whereas I use the ones at the two-digit level for the main analysis. Nevertheless, to ensure a consistent industry classification over time I proceed as follows.

[21] `https://www.scb.se/contentassets/9dd20ce462644cc19f6f04eb2edbbe28/nv0109_kd_slut2022_v1_20240510.pdf`, accessed 26.03.2025.

During the year of the change, I observe both the old and the new industry classifications. For the firms present in the data in the year of the classification change, extending the new industry classification further back in time before the change is straightforward. This way, the industry codes of almost all firms are updated. A firm might be in the data before and after the classification change but not for the year of the change. For these firms, the above method does not work. If the firm appears in the data one year after the classification change, I use the observed classification after the change to update the classification before the change. For firms that are absent for several years around the year of change, I use industry mappings provided by Statistics Sweden. These mappings do not always provide a 1:1 mapping between industries before and after the classification change, so I use the most common transitions for the m:m mappings.

One concern is that changes in the firm structure, e.g., when firms merge with other firms, change the firm ID. To address this concern, I impute changes in firm IDs using worker flows between firms. The auxiliary data set *Registerbaserad Arbetsmarknadsstatistik* (RAMS) contains the universe of employer-employee matches. I impute changes in the firm ID of firms with at least five employees as follows: if more than 50% of the workforce of firm A in year t makes up for more than 50% of the workforce of firm B in year $t+1$, I substitute firm B's firm ID by firm A's firm ID following $t+1$. The empirical results remain unchanged when excluding firms for whom the imputed firm ID differs from the observed firm ID.

Other data sources

- Federal Reserve Economic Data (FRED): Total Factor Productivity at Constant National Prices for Sweden (series RTFPNASEA632NRUG). Accessed Tuesday, January 30, 2024.

B Trends in the Swedish economy

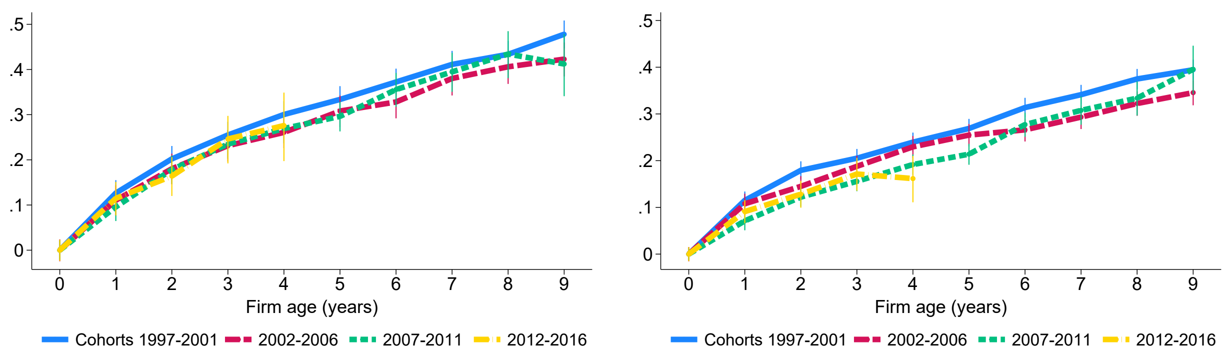
B.1 Trends in firm dynamics

This section documents additional trends in firm size in the Swedish economy. Figure A-1a plots the average physical capital stock at each firm age, normalized by the average capital stock of entrants in logs. The capital stock is defined as fixed assets minus depreciation. The different cohort trajectories overlap, indicating no systematic difference in firms' capital accumulation over time. This suggests that the steepening of the firm size-age relationship documented in the main text is not driven by capital deepening.

Figure A-1: Firm size-age relationship (capital)

(a) Physical capital

(b) Physical capital plus intangibles



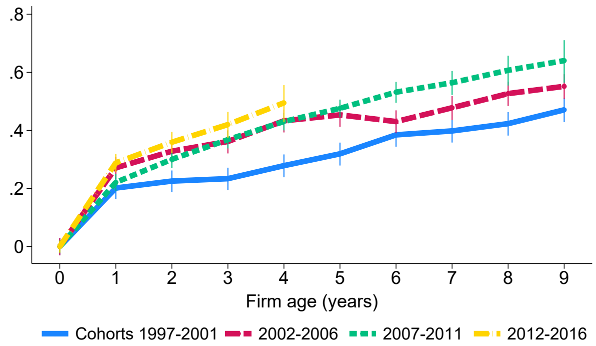
Notes: the figure shows the average capital stock of surviving firms relative to the average capital stock of entrants in logs for different cohorts as indicated in the legend. The left panel captures physical capital measured as fixed assets minus depreciation and the right panel adds intangible assets to the capital measure. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are included.

This picture does not change even when including intangible assets in the measure of the capital stock. Figure A-1b repeats the previous graph for the measure of capital that includes firms' (book value of) intangible capital. The trajectories become slightly more dispersed compared to Figure A-1a; however, there is no systematic trend across cohorts visible.

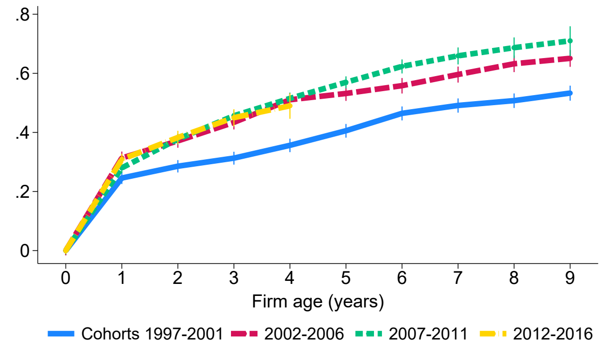
Next, I show the firm size-age relationship for value added and the wage bill by sector. The left column in Figure A-2 is restricted to firms in the goods sector, and the right column to firms in the services sector. The trends are visible in both sectors, particularly in the services sector. That the trends are particularly clear for services firms whose products generally are locally provided and consumed suggests that the trends in firm size are not driven by foreign goods demand.

Figure A-2: Firm size-age relationship (by sector)

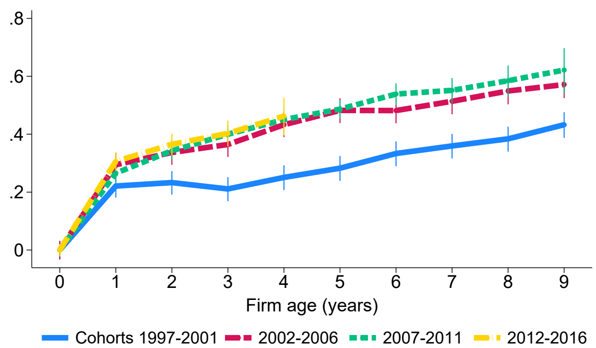
(a) Value added (Goods)



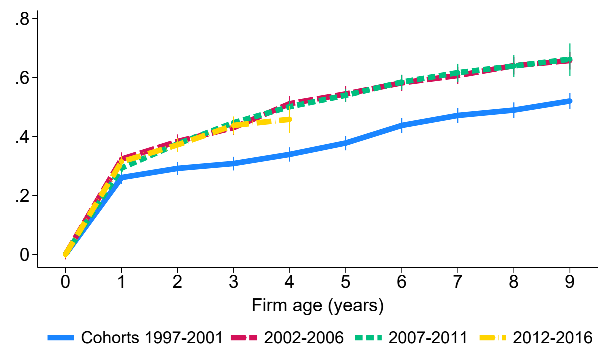
(b) Value added (Services)



(c) Wage bill (Goods)



(d) Wage bill (Services)



Notes: the figure shows the average value added and wage bill of surviving firms relative to the average size of entrants in logs for different cohorts as indicated in the legend. The left column is restricted to firms in the goods sector and the right column to firms in the services sector. 95% confidence bands are included.

Lastly, I assess the robustness of the steepening of the firm-size trajectories using a regression framework. Specifically, I regress firm size (measured by value added or the wage bill) on firm age, controlling for interacted cohort and five-digit industry fixed effects. Formally,

$$Size_{f,t} = \beta_0 + \beta_1 Age_{f,t} + \delta_{c,n} + \epsilon_{f,t}, \tag{A-1}$$

where c and n index cohorts and industries, respectively. I estimate the regression in levels rather than logarithms, since value added can be negative and the wage bill is zero for self-employed entrepreneurs without employees, who are included in the data. Excluding such firm-year observations would introduce selection among firm-size trajectories as entrepreneurs commonly start businesses without any labor costs. To study how the size-age relationship evolves over time, I estimate the regression separately by cohort groups and focus on firms' first five years of life. I restrict attention to cohorts founded before 2013 to ensure each is observed for a full five-year window.

Table A-1 presents the regression results using value added as the measure of firm size. The coefficient on firm age increases noticeably over time. For firms established between 1997 and 1999, each additional year of age is associated, on average, with an increase in value added of 140,468 SEK (in 2017 prices; roughly 14,000 USD). This effect rises to 189,504 SEK for firms founded between 2000 and 2002, 195,526 SEK for those founded between 2003 and 2005, and 193,044 SEK for firms established thereafter. These results confirm not only the steepening of the size–age relationship, but also that this trend was particularly pronounced for cohorts founded around the turn of the millennium. Table A-2 shows very similar patterns when using the wage bill as the size measure.

Table A-1: Size-age regressions: value added

	Size	Size	Size	Size
Age	140,468	189,504	195,526	193,044
	(4,713)	(4,916)	(5,339)	(2,670)
Cohort × industry fixed effects	✓	✓	✓	✓
Cohorts 1997-1999	✓			
Cohorts 2000-2002		✓		
Cohorts 2003-2005			✓	
Cohorts 2006-2012				✓
R^2	0.119	0.148	0.129	0.108
Number of observations	321,027	321,588	212,451	873,447

Notes: the table shows the results of the regression in eq. (A-1) using value added as the firm-size measure. The age coefficient is in units of 2017 SEK (1 SEK ≈ 0.1 USD). All regressions control for interacted cohort and five-digit industry fixed effects. Standard errors are clustered at the firm level. The 2004 cohort is excluded as explained in the main text.

Table A-2: Size-age regressions: wage bill

	Size	Size	Size	Size
Age	79,321	93,508	108,851	109,902
	(2,638)	(2,709)	(3,013)	(1,565)
Cohort × industry fixed effects	✓	✓	✓	✓
Cohorts 1997-1999	✓			
Cohorts 2000-2002		✓		
Cohorts 2003-2005			✓	
Cohorts 2006-2012				✓
R^2	0.122	0.150	0.134	0.115
Number of observations	321,027	321,588	212,451	873,447

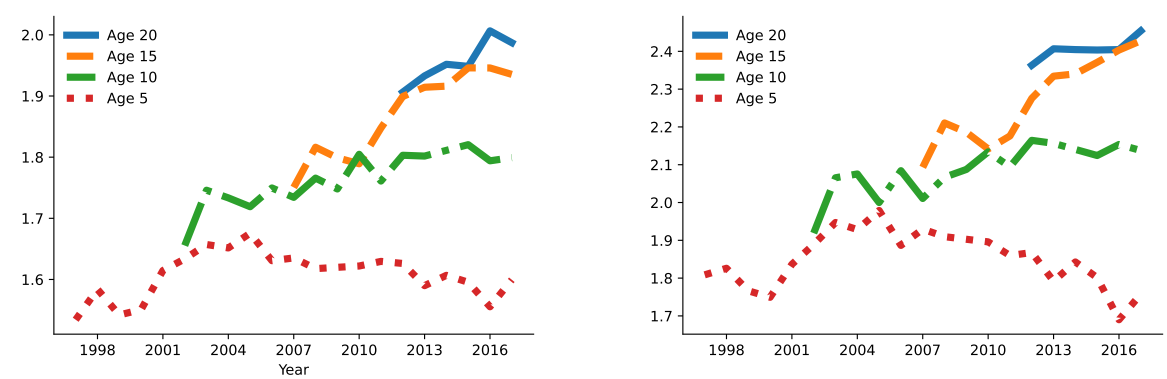
Notes: the table shows the results of the regression in eq. (A-1) using firms' wage bills as the size measure. The age coefficient is in units of 2017 SEK (1 SEK ≈ 0.1 USD). All regressions control for interacted cohort and five-digit industry fixed effects. Standard errors are clustered at the firm level. The 2004 cohort is excluded as explained in the main text.

I further examine the evolution of firm-size dispersion conditional on age. Figure A-3a measures dispersion using the standard deviation of log value added and restricts the sample to firms with positive value added. Figure A-3b instead measures dispersion using the interquartile range of log value added and therefore does not require this restriction. Both figures show that size dispersion remains roughly constant for young firms (age five) while increasing for older firms (age ten and above).

Figure A-3: Size dispersion conditional on firm age

(a) Standard deviation

(b) Interquartile range



Notes: the figures show firm size dispersion conditional on firm age. Dispersion is measured using the standard deviation in the left panel and the interquartile range in the right panel. The firm size measure is log value added.

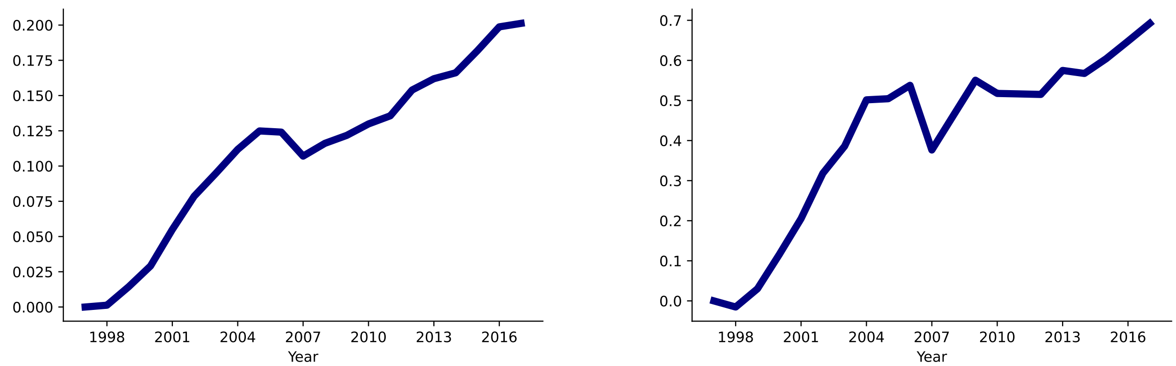
B.2 Trends in the macroeconomy

This section presents the evolution of the two main measures of cross-sectional firm-size dispersion—namely, the standard deviation and interquartile range of log value added—using finer, five-digit industry classifications. The benefit of this approach is that it compares firms operating in more narrowly defined product markets. The drawback is that, at this level of granularity, some industries have a negative 25th percentile of value added, making the logarithm undefined. For the interquartile range in Figure A-4b, I exclude such product groups from the analysis. For the standard deviation in Figure A-4a, all firm-year observations with negative value added are excluded. Despite this limitation, the evolution of firm-size dispersion closely mirrors the patterns documented in the main text, as Figure A-4 illustrates.

Figure A-4: Cross-sectional firm size dispersion

(a) Standard deviation

(b) Interquartile range



Notes: the left panel shows the standard deviation and the right panel the interquartile range of log value added. Both measures are computed within five-digit industries and aggregated using industry labor costs. The value in 1997 is normalized to zero.

C Model

C.1 Solving the dynamic firm problem

The HJB for a high productivity-type firm h reads[22]

$$
\begin{aligned}
&r_t V_t^h(n,[\mu_i],S_t) - \dot{V}_t^h(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^h\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^h\left(n,\left[[\mu_i]_{i\neq k},\mu_k \times \lambda\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^h\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) + (1-S_t) V_t^h\left(n+1,\left[[\mu_i],\lambda\times\varphi^h/\varphi^l\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1} \frac{1}{\psi_I} (I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \Bigg\}
\end{aligned}
$$

The HBJ for a low productivity-type firm l reads

$$
\begin{aligned}
&r_t V_t^l(n,[\mu_i],S_t) - \dot{V}_t^l(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^l\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^l\left(n,\left[[\mu_i]_{i\neq k},\mu_k \times \lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^l\left(n+1,\left[[\mu_i],\lambda\times\varphi^l/\varphi^h\right],S_t\right) + (1-S_t) V_t^l\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1} \frac{1}{\psi_I} (I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \Bigg\}.
\end{aligned}
$$

I solve for the value function of a high-type firm, however the steps for the low-type firm are equivalent. For clarity, I suppress the dependence of the value function on S_t in the following. Guess that the value function of the firm consists of a component that is common to all lines and a line-specific component

$$V_t^h(n,[\mu_i]) = V_{t,P}^h(n) + \sum_{k=1}^{n} V_{t,M}^h(\mu_k).$$

[22] The derivation follows Peters (2020) when applicable.

Substituting the guess into the HJB, $V^h_{t,P}(n)$ and $V^h_{t,M}(\mu_k)$ solve the following differential equations

$$\begin{aligned} r_t V^h_{t,M}(\mu_i) - \dot{V}^h_{t,M}(\mu_i) =& \pi(\mu_i) - \tau_t V^h_{t,M}(\mu_i) \\ &+ \max_{I_i}\left\{ I_i\left[V^h_{t,M}(\mu_i \times \lambda) - V^h_{t,M}(\mu_i)\right] - w_t \mu_i^{-1}\frac{1}{\psi_I}(I_i)^{\zeta}\right\} \end{aligned} \tag{A-2}$$

and

$$\begin{aligned} r_t V^h_{t,P}(n) - \dot{V}^h_{t,P}(n) &= \sum_{k=1}^{n} \tau_t \left[V^h_{t,P}(n-1) - V^h_{t,P}(n)\right] \\ + \max_{[x_k]} &\left\{ \sum_{k=1}^{n} x_k \left[V^h_{t,P}(n+1) - V^h_{t,P}(n) + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l)\right] - w_t \frac{1}{\psi_x}(x_k)^{\zeta}\right\}. \end{aligned} \tag{A-3}$$

Assume that in steady-state $V^h_{t,P}$ and $V^h_{t,M}$ grow at the constant rate g. Using this guess in eq. (A-2) and following Peters (2020), we obtain for $V^h_{t,M}(\mu_i)$

$$V^h_{t,M}(\mu_i) = \frac{\pi(\mu_i) + \frac{\zeta-1}{\psi_I}(I_i)^{\zeta} w_t \mu_i^{-1}}{\rho+\tau},$$

where I_i solves

$$I_i = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta-1}{\psi_I}(I_i)^{\zeta}\right)\left(1-\frac{1}{\lambda}\right)\frac{\psi_I}{\zeta(\rho+\tau)}\right)^{\frac{1}{\zeta-1}}. \tag{A-4}$$

Eq. (A-4) shows that internal innovation rates I_i are time invariant, and independent of the product line and the productivity type of the firm, $I \equiv I^h = I^l$.

With this at hand, we can turn back to the differential equation for $V^h_{t,P}(n)$ in eq. (A-3). In addition to the guess that $V^h_{t,P}(n)$ grows at rate g, conjecture that $V^h_{t,P}(n) = n \times v^h_t$. Combined with the Euler we get

$$(\rho+\tau) n v^h_t = \max_{[x_k]}\left\{\sum_{k=1}^{n} x_k \left[v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l)\right] - w_t \frac{1}{\psi_x}(x_k)^{\zeta}\right\}. \tag{A-5}$$

The optimality condition for x_k is given by

$$v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) = w_t \frac{\zeta}{\psi_x}(x_k)^{\zeta-1}. \tag{A-6}$$

Several observations are noteworthy. First, eq. (A-6) shows that optimal expansion rates are independent of quality and productivity gaps in line k. We can hence drop the item indexation: $x_k = x^d$, where $d \in \{h, \ell\}$. Second, $v_t, V^h_{t,M}, w_t$ all grow at the same rate g, which implies that expansion rates are constant over time. We can hence write eq. (A-5) as

$$v_t^h = \frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^h)^\zeta w_t.$$

Gathering all terms, the value function is given by

$$\begin{aligned} V_t^h(n,[\mu_i]) =& V^h_{t,P}(n) + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n v_t^h + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n\frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^h)^\zeta w_t + \sum_{k=1}^{n}\frac{\pi(\mu_k)+\frac{\zeta-1}{\psi_I}I^\zeta w_t \mu_k^{-1}}{\rho+\tau}, \end{aligned} \tag{A-7}$$

which is the expression for the value function stated in the main text, Proposition 1.

Using the expression for v_t^h, write the optimality condition in eq. (A-6) as

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right) + (1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^l}{\varphi^h}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1}.$$

Following the same steps for low-productivity firms, we obtain the optimality condition

$$\frac{\zeta-1}{\psi_x}(x^l)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^h}{\varphi^l}\right) + (1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta-1}.$$

Lastly, I prove that more productive firms choose higher expansion R&D rates, i.e., $x^h > x^l$. Intuitively, the proof shows that an increase in productivity raises the stream of profits in a product line. The continuation value and the marginal cost of expansion R&D have to rise for the optimality condition of the expansion R&D rate to hold, implying that the expansion R&D rate increases. First note that product markups are increasing in firm productivity, as shown in eq. (4). Next, I totally differentiate the optimality condition for the expansion

R&D rate and show that the expansion R&D rate is increasing in the markup.

Write the optimality condition for expansion R&D as[23]

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^h}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^h}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^l}\right)$$
$$=(\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1},$$

where μ^h and μ^l denote the initial markup charged when facing a high- and low-productivity firm. Totally differentiate with respect to markups and the expansion R&D rate

$$S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left((\rho+\tau)(x^h)^{\zeta-2}-(x^h)^{\zeta-1}\right)dx^h,$$

where $d\mu^h = d\mu^l \equiv d\mu$. Since $x^h > 0$, the above can be rearranged to

$$\frac{1}{(x^h)^{\zeta-2}}\left[S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)\right]d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left(\rho+\tau-x^h\right)dx^h.$$

The left hand side captures the effect of changes in markups on profits. The right hand side captures the effect of changes in the expansion R&D rate on the continuation value and marginal costs of expansion R&D. From eq. (A-4) we know that $\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta > 0$, otherwise the optimal internal R&D rate is negative. For a stationary firm size distribution, we must further have $\tau > x^h$. From this, it follows that $dx^h/d\mu > 0$, which concludes the proof.

C.2 Joint distribution of quality and productivity gaps

The distribution of quality and productivity gaps between incumbent and laggard firms characterizes economic aggregates in the model. On the one hand, quality and productivity gaps define product markups that determine labor demand. On the other hand, the joint distribution characterizes the share of product lines operated by each productivity type, which is a state variable in the firm's optimization problem. This section derives the joint distribution of quality and productivity gaps as a function of firm policies, which allows the equilibrium distribution to be solved jointly with the policies.

[23]This uses the optimality condition of the high-type firm but the one of the low type works equivalently.

The distribution of quality and productivity gaps across product lines is characterized by a set of infinitely many differential equations. For simplicity, I characterize the differential equations for firm-type specific expansion R&D rates, x_t^h and x_t^ℓ, and homogeneous internal R&D rates, I_t, as proven in Proposition 1 for a balanced growth path.[24] The measure ν_t of product lines in which an incumbent of productivity type φ_f is Δ quality steps ahead of a $\varphi_{f'}$ type laggard follows (for $\Delta \geq 2$)

$$\dot{\nu}_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right) = I_t \nu_t\left(\Delta - 1, \frac{\varphi_f}{\varphi_{f'}}\right) - \nu_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right)(I_t + \tau_t). \tag{A-8}$$

For product lines where the incumbent is one step ahead ($\Delta = 1$), this measure follows

$$\begin{aligned}
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^h}\right) &= (1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t - \nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^l}\right) &= (1 - S_t)x_t^l(1 - S_t) + z_t(1 - p^h)(1 - S_t) - \nu_t\left(1, \frac{\varphi^l}{\varphi^l}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^h}\right) &= S_t x_t^h S_t + z_t p^h S_t - \nu_t\left(1, \frac{\varphi^h}{\varphi^h}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^l}\right) &= S_t x_t^h(1 - S_t) + z_t p^h(1 - S_t) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right)(I_t + \tau_t).
\end{aligned} \tag{A-9}$$

Changes in ν_t are due to inflows and outflows. Outflows from state Δ arise from successful internal R&D (quality gap increases from Δ to $\Delta + 1$) and creative destruction (Δ is reset to unity), as captured by the terms following the minus signs. Inflows vary with the quality gap. For $\Delta \geq 2$, inflows into state Δ are due to successful internal R&D in product lines with quality gaps of $\Delta - 1$. For $\Delta = 1$, inflows result from creative destruction. For example, the measure of products with a low-type incumbent and high-type laggard $\nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)$ increases due to low-type incumbents and entrants replacing high-type incumbents, captured by $(1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t$ in eq. (A-9).

The following characterizes the two-dimensional distribution of quality and productivity gaps along the BGP. I solve for the steady state distribution over quality and productivity gaps by setting the differential equations characterizing the law-of-motion in eq. (A-8) and (A-9) equal to zero. From this, one obtains the stationary mass of product lines with quality gap

[24] The distribution along the transition path is characterized in the Appendix, Section D.

λ^{Δ} and productivity gap φ^i/φ^j

$$\nu\left(\Delta, \frac{\varphi^l}{\varphi^h}\right) = \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{(1-S)x^l S + z(1-p^h)S}{I}$$
$$\nu\left(\Delta, \frac{\varphi^l}{\varphi^l}\right) = \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{(1-S)x^l(1-S) + z(1-p^h)(1-S)}{I}$$
$$\nu\left(\Delta, \frac{\varphi^h}{\varphi^h}\right) = \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{Sx^h S + zp^h S}{I}$$
$$\nu\left(\Delta, \frac{\varphi^h}{\varphi^l}\right) = \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{Sx^h(1-S) + zp^h(1-S)}{I}.$$

Summing over all Δ for a given productivity gap gives $S_{\varphi^l,\varphi^h}, S_{\varphi^l,\varphi^l}, S_{\varphi^h,\varphi^h}, S_{\varphi^h,\varphi^l}$ as stated in Proposition 1 in the main text. It follows that

$$\Pr\left(\Delta \le d, \frac{\varphi^l}{\varphi^h}\right) = \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right)$$
$$\Pr\left(\Delta \le d, \frac{\varphi^l}{\varphi^l}\right) = \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^l}\right) = S_{\varphi^l,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right)$$
$$\Pr\left(\Delta \le d, \frac{\varphi^h}{\varphi^h}\right) = \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^h}\right) = S_{\varphi^h,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right)$$
$$\Pr\left(\Delta \le d, \frac{\varphi^h}{\varphi^l}\right) = \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^l}\right) = S_{\varphi^h,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^d\right).$$

Focusing on product lines where a low-productivity incumbent faces a high-productivity second-best firm:

$$P\left(\Delta \le d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-d[\ln(I+\tau)-\ln I]}\right)$$

or

$$P\left(\ln\left(\lambda^{\Delta}\right) \le d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}d}\right).$$

Conditional on the productivity gap, $\ln\left(\lambda^{\Delta}\right)$ is exponentially distributed with parameter

$\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Further

$$P\left(\lambda^{\Delta} \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - d^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter $\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Denote $\theta = \frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. We then have

$$P\left(\lambda^{\Delta} \leq m, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - m^{-\theta}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter θ. As in Peters (2020), θ is affected by the rate of internal R&D I relative to creative destruction τ. The higher the rate of internal R&D, the more mass is in the tail of the quality gap distribution. The difference to Peters (2020) is that, in this model, quality gaps *conditional* on the productivity gap are Pareto distributed.

The following uses the obtained results to solve for the aggregate labor income share, the misallocation measure and the aggregate markup. After repeating the same steps for lines with different productivity gaps, we obtain the aggregate labor income share as follows[25]

$$\begin{aligned}\Lambda &= \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{1}{\varphi_k/\varphi_n}\frac{1}{m}S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\ &= \frac{\theta}{\theta+1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n}.\end{aligned} \tag{A-10}$$

The TFP misallocation statistic $\mathcal{M}$ is then given by

$$\begin{aligned}\mathcal{M} &= \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int\left[\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}\right]dm}}{\Lambda} \\ &= \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\left[S_{\varphi_k,\varphi_n}\left(\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\right)-\frac{1}{\theta}\right)\right]}}{\Lambda},\end{aligned}$$

where I have made use of

$$\int_1^{\infty}\ln\left(\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm = \left[\frac{\theta\ln(m)+1}{\theta m^{\theta}} + C\right]_1^{\infty} = -\frac{1}{\theta}.$$

[25] For the derivation, I assume a continuous distribution of quality gaps.

Alternatively note that this expression is equal to $-S_{\varphi_k,\varphi_n}E[\ln(\lambda^{\Delta})|\varphi_k,\varphi_n]$. I have shown above that $\ln(\lambda^{\Delta})$ conditional on the productivity gap is exponentially distributed with parameter θ. From the characteristics of an exponential distribution, its expected value is $1/\theta$.

The aggregate markup is then given by

$$\begin{aligned} E[\mu] =& \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{\varphi_k}{\varphi_n}mS_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\ =&\frac{\theta}{\theta-1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{\varphi_k}{\varphi_n}S_{\varphi_k,\varphi_n}. \end{aligned} \tag{A-11}$$

This concludes the derivations of the aggregate labor income share, TFP misallocation statistic and aggregate markup. In Peters (2020), the step size of innovation and the rate of creative destruction relative to internal R&D, captured by θ, fully characterize the aggregate labor income share, the misallocation measure, and the aggregate markup. In this model, these statistics further depend on the size and distribution of the productivity gaps. For example, a rise in the productivity gap or a reallocation of sales shares towards high-productivity firms lowers the aggregate labor income share in eq. (A-10) and raises the aggregate markup in eq. (A-11), ceteris paribus.

C.3 Deriving the steady-state growth rate of aggregate variables

The growth rate of Q_t determines the growth rate of aggregate variables

$$g = \frac{\dot{Q}_t}{Q_t} = \frac{\partial \ln(Q_t)}{\partial t}.$$

Quality of a product in a given product line increases through internal R&D, expansion R&D or firm entry. For the growth rate of Q_t over a discrete time interval Δ, we have

$$\ln(Q_{t+\Delta}) = \int_0^1 \Big[(\Delta I + \Delta S x^h + \Delta(1-S)x^l + \Delta z)\ln(\lambda) + \ln(q_{t,i})\Big]\, di$$

so that

$$\frac{\ln(Q_{t+\Delta}) - \ln(Q_t)}{\Delta} = \Big(I + Sx^h + (1-S)x^l + z\Big)\ln(\lambda).$$

For $\Delta \to 0$, $g = \Big(I + Sx^h + (1-S)x^l + z\Big)\ln(\lambda)$ as stated in Proposition 1.

C.4 Solving for the steady state equilibrium

In the model, there are the seven unknown variables $x^h, x^l, I, z, \tau, \frac{Y_t}{w_t}, S$ and the markup distribution $\nu()$ in seven equations plus the system of differential equations characterizing $\nu()$.

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta - 1}{\psi_I} I^{\zeta} \right) \left(1 - \frac{1}{\lambda} \right) \frac{\psi_I}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x} (x^h)^{\zeta} + S \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda} \right) + \frac{\zeta - 1}{\psi_I} I^{\zeta} \lambda^{-1} \right) + (1 - S) \left(\frac{Y_t}{w_t} \left(1 - \frac{\varphi^l}{\varphi^h} \frac{1}{\lambda} \right) + \frac{\zeta - 1}{\psi_I} I^{\zeta} \lambda^{-1} \frac{\varphi^l}{\varphi^h} \right) \\ = (\rho + \tau) \frac{\zeta}{\psi_x} (x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x} (x^l)^{\zeta} + S \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda} \frac{\varphi^h}{\varphi^l} \right) + \frac{\zeta - 1}{\psi_I} I^{\zeta} \lambda^{-1} \frac{\varphi^h}{\varphi^l} \right) + (1 - S) \left(\frac{Y_t}{w_t} \left(1 - \frac{1}{\lambda} \right) + \frac{\zeta - 1}{\psi_I} I^{\zeta} \lambda^{-1} \right) \\ = (\rho + \tau) \frac{\zeta}{\psi_x} (x^l)^{\zeta - 1}$$

Free entry condition

$$p^h \Big(S V_t^h(1, \lambda) + (1 - S) V_t^h(1, \lambda \times \varphi^h / \varphi^l) \Big) + (1 - p^h) \Big(S V_t^l(1, \lambda \times \varphi^l / \varphi^h) + (1 - S) V_t^l(1, \lambda) \Big) = \frac{1}{\psi_z} w_t,$$

where

$$V_t^d(1, \mu) = \frac{1}{(\rho + \tau)} \frac{\zeta - 1}{\psi_x} (x^d)^{\zeta} w_t + \frac{Y_t \left(1 - \frac{1}{\mu} \right) + \frac{\zeta - 1}{\psi_I} I^{\zeta} w_t \mu^{-1}}{\rho + \tau}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t} \sum_{\frac{\varphi_j}{\varphi_{j'}}} \sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}} \nu \left(i, \frac{\varphi_j}{\varphi_{j'}} \right) + \frac{1}{\psi_I} I^{\zeta} \sum_{\frac{\varphi_j}{\varphi_{j'}}} \sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}} \nu \left(i, \frac{\varphi_j}{\varphi_{j'}} \right) + S \frac{1}{\psi_x} (x^h)^{\zeta} + (1 - S) \frac{1}{\psi_x} (x^l)^{\zeta} + \frac{z}{\psi_z}$$

Creative destruction

$$\tau = z + S x^h + (1 - S) x^l$$

Share of high productivity type

$$S=\sum_{i=1}^{\infty}\left[\nu\left(i,\frac{\varphi^h}{\varphi^h}\right)+\nu\left(i,\frac{\varphi^h}{\varphi^l}\right)\right],$$

where ν, the stationary distribution of quality and productivity gaps, is characterized by

$$0=\dot{\nu}\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)=I\nu\left(\Delta-1,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)(I+\tau)\quad\text{for}\quad\Delta\geq 2$$

and for the case of a unitary quality gap

$$\begin{aligned}
0=\dot{\nu}\left(1,\frac{\varphi^l}{\varphi^h}\right)&=(1-S)x^lS+z_t(1-p^h)S-\nu\left(1,\frac{\varphi^l}{\varphi^h}\right)(I+\tau)\\
0=\dot{\nu}\left(1,\frac{\varphi^l}{\varphi^l}\right)&=(1-S)x^l(1-S)+z_t(1-p^h)(1-S)-\nu\left(1,\frac{\varphi^l}{\varphi^l}\right)(I+\tau)\\
0=\dot{\nu}\left(1,\frac{\varphi^h}{\varphi^h}\right)&=Sx^hS+z_tp^hS-\nu\left(1,\frac{\varphi^h}{\varphi^h}\right)(I+\tau)\\
0=\dot{\nu}\left(1,\frac{\varphi^h}{\varphi^l}\right)&=Sx^h(1-S)+z_tp^h(1-S)-\nu\left(1,\frac{\varphi^h}{\varphi^l}\right)(I+\tau).
\end{aligned}$$

To simplify the system of equations, first rewrite the rate of creative destruction

$$z=(\tau-Sx^h-(1-S)x^l)$$

such that z can be substituted out from the remaining equations. Second, based on Proposition 1, we know

$$S=\frac{Sx^h+zp^h}{\tau}.$$

Third, the optimality conditions for expansion rates (multiplied by p^h and $(1-p^h)$) and the free entry condition together imply

$$\frac{1}{\psi_x}p^h(x^h)^{\zeta-1}+\frac{1}{\psi_x}(1-p^h)(x^l)^{\zeta-1}=\frac{1}{\psi_z\zeta}.$$

The system of equilibrium conditions can hence be reduced to:

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} \psi_I - (\zeta - 1) I^{\zeta} \right) \frac{\left(1 - \frac{1}{\lambda}\right)}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\begin{aligned}\frac{\zeta - 1}{\psi_x}(x^h)^{\zeta} + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{1}{\psi_I}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{\varphi^l}{\psi_I \varphi^h}\right) \\ = (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}\end{aligned}$$

Optimality condition for low-productivity type expansion rate

$$\begin{aligned}\frac{\zeta - 1}{\psi_x}(x^l)^{\zeta} + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{\varphi^h}{\psi_I \varphi^l}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{1}{\psi_I}\right) \\ = (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}\end{aligned}$$

Free entry

$$p^h \frac{(x^h)^{\zeta - 1}}{\psi_x} + (1 - p^h)\frac{(x^l)^{\zeta - 1}}{\psi_x} = \frac{1}{\psi_z \zeta}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t}\Lambda + \Lambda_I + S\frac{1}{\psi_x}(x^h)^{\zeta} + (1 - S)\frac{1}{\psi_x}(x^l)^{\zeta} + \frac{\tau - Sx^h - (1 - S)x^l}{\psi_z},$$

where

$$\begin{aligned}\Lambda =& \frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n} \\ \Lambda_I =& \frac{1}{\psi_I}I^{\zeta}\frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n} \\ \theta =& \frac{\ln(I + \tau) - \ln(I)}{\ln(\lambda)}\end{aligned}$$

Sales share of high productivity type

$$S = \frac{Sx^h + (\tau - Sx^h - (1 - S)x^l)p^h}{\tau}$$

The expressions related to the labor market clearing condition are derived in Section C.2. This constitutes a system of seven equations in seven unknowns ($x^h, x^l, I, \tau, \frac{Y_t}{w_t}, S$), which I solve using a root finder.

C.5 Firm markups

Firm markups are defined by $\mu_f = \frac{py_f}{wl_f} = \left(\frac{1}{n}\sum_{k=1}^{n}\mu_{kf}^{-1}\right)^{-1}$. Therefore

$$\ln\mu_f = -\ln\left(\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1}\right).$$

Rewrite the term in brackets (for a high-productivity firm) as

$$\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1} = \frac{1}{n}\sum_{k=1}^{n}e^{-\ln\mu_k} = \frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right), \tag{A-12}$$

where i indexes the product lines where the high productivity firm faces a low productivity second best producer, j the lines where it faces a high productivity second best producer and $n_i + n_j = n$. A two-dimensional linear Taylor expansion around $\ln\lambda = 0$ and $\ln\frac{\varphi^h}{\varphi^l} = 0$ gives

$$\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right) \approx 1 - \left(\frac{1}{n}\sum_{k=1}^{n}\Delta_k\right)\ln\lambda - \frac{n_i}{n}\ln\left(\frac{\varphi^h}{\varphi^l}\right)$$

such that

$$E\left[\ln\mu_f|\text{firm age} = a_f, \varphi^h\right] \approx E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda + (1-S)\ln\left(\frac{\varphi^h}{\varphi^l}\right),$$

where I have used the fact that (in expectation) the share of the firm's product lines with a low productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the low productivity type. From Peters (2020), we know that

$$E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda = \left(1 + I\times E[a_P^h|a_f]\right)\ln\lambda,$$

where $E[a_P^h|a_f]$ denotes the average product age of a high-productivity type firm conditional on firm age a_f and

$$E[a_P^h|a_f] = \frac{1}{x^h}\left(\frac{\frac{1}{\tau}\left(1-e^{-\tau a_f}\right)}{\frac{1}{x^h+\tau}\left(1-e^{-(x^h+\tau)a_f}\right)} - 1\right)\left(1-\phi^h(a_f)\right) + a_f\phi^h(a_f)$$

$$
\begin{aligned}
\phi^h(a) &= e^{-x^h a} \frac{1}{\gamma^h(a)} \ln \left(\frac{1}{1-\gamma^h(a)} \right) \\
\gamma^h(a) &= \frac{x^h \left(1 - e^{-(\tau - x^h)a} \right)}{\tau - x^h e^{-(\tau - x^h)a}}.
\end{aligned}
$$

For a firm of the low-productivity type, the last term in eq. (A-12) reads

$$
\frac{1}{n} \left(\sum_{i=1}^{n_i} e^{-\ln \Delta_i \ln \lambda} + \sum_{j=1}^{n_j} e^{-\ln \frac{\varphi^l}{\varphi^h} - \ln \Delta_j \ln \lambda} \right),
$$

where i indexes the product lines where the low-productivity producer faces a low-productivity second best producer, j the lines where it faces a high-productivity second best producer and $n_i + n_j = n$. Following the same steps as for a high-productivity firm, this time linearizing around $\ln \frac{\varphi^l}{\varphi^h} = 0$ (and $\ln \lambda = 0$) gives

$$
E \left[\ln \mu_f | \text{firm age} = a_f, \varphi^l \right] \approx \left(1 + I \times E[a_P^l | a_f] \right) \ln \lambda + S \ln \left(\frac{\varphi^l}{\varphi^h} \right),
$$

where again I have made use of the fact that (in expectation) the share of the firm's product lines with a high-productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the high-productivity type. $E[a_P^l | a_f]$ is exactly defined as $E[a_P^h | a_f]$ with x^h replaced by x^l in the above expressions.

C.6 Markup-age relationship

As show in the Appendix, Section C.5 the expected log markup conditional on firm age a_f for a high-productivity type firm is

$$
E \left[\ln \mu_f | a_f, \varphi^h \right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^h | a_f] \right)}_{\text{Quality improvements}} + \underbrace{(1 - S) \times \ln \left(\frac{\varphi^h}{\varphi^l} \right)}_{\text{Productivity advantage}}, \tag{A-13}
$$

where $E[a_P^h | a_f]$ is the average product age of a high-type firm conditional on firm age.

The expected firm markup conditional on age consists of two terms. The first term in eq. (A-13) is akin to Peters (2020) and reflects that internal R&D at the product level translates into markup growth at the firm level as it ages. In Peters (2020), this term holds for all firms, whereas in this model, this term is specific to the productivity type of the firm, as the average product age varies by firm type. The second term in eq. (A-13) captures a new level effect that heterogeneity in productivity introduces. The intuition is that if a high-type incumbent faces a low-type second-best firm, it can charge a φ^h/φ^l higher markup, which occurs in expectation in $1-S$ of the incumbent's product lines. The markup-age relationship for high-productivity firms then equals $E\left[\ln \mu_f | a_f, \varphi^h\right] - E\left[\ln \mu_f | 0, \varphi^h\right]$.

The expected markup conditional on firm age for a low-productivity type firm follows

$$E\left[\ln \mu_f | a_f, \varphi^l\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^l | a_f]\right)}_{\text{Quality improvements}} + \underbrace{S \times \ln\left(\frac{\varphi^l}{\varphi^h}\right)}_{\text{Productivity disadvantage}}. \tag{A-14}$$

The first term captures quality improvements through internal R&D. The second term in eq. (A-14) differs from eq. (A-13). Low-productivity incumbents face a high-productivity second-best firm in a share S of their product lines. Since $\varphi^l < \varphi^h$, this term is negative. The markup-age relationship is then defined likewise as for high-productivity firms.

C.7 Firm size distribution

The mass of high- and low-productivity type firms with $n \geq 2$ products follows the differential equations

$$\begin{aligned} \dot{M}_t^h(n) &= (n-1)x_t^h M_t^h(n-1) + (n+1)\tau_t M_t^h(n+1) - n(x_t^h + \tau_t)M_t^h(n) \\ \dot{M}_t^l(n) &= (n-1)x_t^l M_t^l(n-1) + (n+1)\tau_t M_t^l(n+1) - n(x_t^l + \tau_t)M_t^l(n), \end{aligned} \tag{A-15}$$

whereas the mass of firms with one product evolves according to

$$\begin{aligned} \dot{M}_t^h(1) &= z_t p^h + 2\tau_t M_t^h(2) - (x_t^h + \tau_t)M_t^h(1) \\ \dot{M}_t^l(1) &= z_t(1-p^h) + 2\tau_t M_t^l(2) - (x_t^l + \tau_t)M_t^l(1). \end{aligned} \tag{A-16}$$

The mass of firms with n products increases through firms with $n-1$ products expanding to size n at rate x_t^h or x_t^l per product or through firms with $n+1$ products losing a product at the rate of aggregate creative destruction τ_t. The mass of firms with n products decreases through firms with n products either gaining or losing a product through expansion or creative destruction. The mass of firms with one product additionally increases through firm entry.

Proposition A-1. *The stationary firm size distribution along the balanced growth path is characterized as follows.*

1. *The mass of high and low productivity firms with n products is*

$$M^h(n) = \frac{(x^h)^{n-1} z p^h}{n \tau^n} = \frac{z p^h}{x^h} \frac{1}{n} \left(\frac{x^h}{\tau} \right)^n$$
$$M^l(n) = \frac{(x^l)^{n-1} z (1 - p^h)}{n \tau^n} = \frac{z (1 - p^h)}{x^l} \frac{1}{n} \left(\frac{x^l}{\tau} \right)^n.$$

2. *The total mass of firms with n products is*

$$M(n) = M^h(n) + M^l(n) = \frac{(x^h)^{n-1} z p^h + (x^l)^{n-1} z (1 - p^h)}{n \tau^n}.$$

3. *The mass of firms of each productivity type is*

$$M^h = \sum_{n=1}^{\infty} M^h(n) = \frac{z p^h}{x^h} \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{x^h}{\tau} \right)^n = \frac{z p^h}{x^h} \ln \left(\frac{\tau}{\tau - x^h} \right)$$
$$M^l = \sum_{n=1}^{\infty} M^l(n) = \frac{z (1 - p^h)}{x^l} \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{x^l}{\tau} \right)^n = \frac{z (1 - p^h)}{x^l} \ln \left(\frac{\tau}{\tau - x^l} \right)$$

4. *The total mass of firms is*

$$M = M^h + M^l.$$

Proof. These results follow from setting the time derivatives in equations (A-15) and (A-16) equal to zero and solving the system of equations. □

For each firm type, the share of firms with n products, $M^h(n)/M^h$ and $M^l(n)/M^l$, follows the PDF of a logarithmic distribution with parameter x^h/τ and x^l/τ as in Lentz and Mortensen (2008). The firm size distribution is highly skewed to the right.

Since there is a continuum of mass one of products and each product is mapped to one firm $\sum_{i=1}^{\infty} M(n) \times n = 1$. Further, the mass of high-productivity type firms producing n products is related to the share of product lines operated by high-type firms, S, as follows

$$S = \sum_{n=1}^{\infty} M^h(n) \times n = \frac{z p^h}{\tau - x^h}.$$

D Computation of transition dynamics

In this section, I lay out the numerical procedure to solve for the transition path. Since time is continuous, I solve a discretized version of the model where the solution converges to the one in continuous time for small enough time intervals. As shown in Appendix C, value functions are additive across product lines. Therefore, I solve the problem of two representative one-product firms: one of the high productivity type and one of the low productivity type.

I normalize the value function by the wage w_t to obtain a stationary problem. The value function for the high-type firm (in discrete time) reads

$$\begin{aligned}
\frac{V_t^h(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^h}\left\{x_t^h \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^h(1,\lambda,S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^h(1,\lambda\frac{\varphi^h}{\varphi^l},S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^h)^\zeta dt\right\} \\
&+ \max_{I_t^h}\left\{I_t^h \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^h(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-17}$$

The value function for the low-type firm reads

$$\begin{aligned}
\frac{V_t^l(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^l}\left\{x_t^l \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^l(1,\lambda\frac{\varphi^l}{\varphi^h},S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^l(1,\lambda,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^l)^\zeta dt\right\} \\
&+ \max_{I_t^l}\left\{I_t^l \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^l(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-18}$$

From this, one obtains the first order conditions for the policy functions. For the optimal expansion R&D rate of the high type firm x_t^h (again suppressing the dependence of the value function on S_t):

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^h_{t+dt}(1,\lambda)}{w_{t+dt}}+(1-S_{t+dt})\frac{V^h_{t+dt}(1,\lambda\frac{\varphi^h}{\varphi^l})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^h_t)^{\zeta-1} \tag{A-19}$$

and for the low type firm x^l_t:

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^l_{t+dt}(1,\lambda\frac{\varphi^l}{\varphi^h})}{w_{t+dt}}+(1-S_{t+dt})\frac{V^l_{t+dt}(1,\lambda)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^l_t)^{\zeta-1}. \tag{A-20}$$

Both are independent of the markup μ_i. For the optimal internal R&D rates of the high type, I^h_t, one obtains

$$\exp(-r_{t+dt}dt)\left(\frac{V^h_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^h_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^h_t)^{\zeta-1} \tag{A-21}$$

and similarly for I^l_t

$$\exp(-r_{t+dt}dt)\left(\frac{V^l_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^l_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^l_t)^{\zeta-1}. \tag{A-22}$$

Equations (A-17) to (A-22) characterize the firm problem in discrete time. These equations are supplemented by the law of motion for the two dimensional distribution of quality and productivity gaps

$$\nu_{t+dt}\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)=dt\left[I_{\mu_i,t}\nu_t\left(\Delta-1,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)(I_{\mu_i,t}+\tau_t)\right]\quad\text{for}\quad\Delta\geq 2 \tag{A-23}$$

and for product lines with a unitary quality gap, $\Delta = 1$,

$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)=dt\left[(1-S_t)x^l_tS_t+z_t(1-p^h)S_t-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^l}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)=dt\left[(1-S_t)x^l_t(1-S_t)+z_t(1-p^h)(1-S_t)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^h}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)=dt\left[S_tx^h_tS_t+z_tp^hS_t-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$

$$\nu_{t+dt}\left(1, \frac{\varphi^h}{\varphi^l}\right) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right) = dt\left[S_t x_t^h(1-S_t) + z_t p^h(1-S_t) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right)(I_{\mu_i,t} + \tau_t)\right] \tag{A-24}$$

and a standard Euler equation

$$\frac{C_{t+dt}}{C_t} = \exp(-\rho dt)(1 + r_{t+dt}dt). \tag{A-25}$$

Further, the (static) free entry and labor market clearing conditions remain unchanged and are characterized in the main text by equations (10) and (12).

The algorithm to compute the transition path assumes that an initial and ending balanced growth path have been solved for including the (stationary) two-dimensional distribution of quality and productivity gaps. I choose $dt = 0.02$ and set the transition period to 300 years (T), after which I assume the economy has reached its new balanced growth path. I further truncate the two dimensional distribution of quality and productivity gaps along the quality dimension at $\Delta = 30$, implying a maximum quality gap of λ^{30}. No mass reaches this state during the transition such that this assumption is satisfied. I then compute the transition path as follows:

1. Guess a path of wage growth $\frac{w_{d+dt}}{w_t}$ and income to wage ratios Y_t/w_t over the transition (equal to their values in the final balanced growth path)
   (a) Guess a path for S_t over the transition (equal to its value in the final balanced growth path).
      i. Starting backwards in period T, solve for optimal policy functions in $T - dt$ using equations (A-19)-(A-22).[26]
      ii. Solve for τ_{T-dt} that ensures that the free entry condition (10) holds.
      iii. Compute the value function in $T - dt$ using equations (A-17) and (A-18).
      iv. Iterate backwards until the first time period.

[26] I solve for the optimal I_i (at each point in time) over a two-dimensional grid of quality and productivity gaps.

v. Starting from the initial balanced growth path, simulate S_t forward using[27]

$$S_{t+dt} = S_t + dt \left[S_t x_t^h (1 - S_t) - (1 - S_t) x_t^l S_t + z_t (p^h (1 - S_t) - (1 - p^h) S_t) \right],$$

where z_t can be substituted out by equation (9).

(b) Update the guess for S_t from step v using bisection and go back to step i. Iterate until the guessed path for S_t converges to the implied one.

2. Starting from the initial balanced growth path, simulate the two dimensional distribution of quality and productivity gaps forward using equations A-23 and A-24.

3. Solve for the implied sequence of $\frac{Y_t}{w_t}$ from the labor market clearing condition.

4. Compute the sequence of quality growth using

$$\frac{Q_{t+dt}}{Q_t} = \exp\left(\left[\int_0^1 I_{\mu_i, t+dt} di + S_{t+dt} x_{t+dt}^h + (1 - S_{t+dt}) x_{t+dt}^l + z_{t+dt} \right] dt \ln(\lambda) \right).$$

5. Compute the sequence of aggregate productivity growth using

$$\frac{\Phi_{t+dt}}{\Phi_t} = \left(\frac{\varphi^h}{\varphi^l} \right)^{S_{t+dt} - S_t}.$$

6. Using the two dimensional distribution of quality and productivity gaps, compute the sequence of $\mathcal{M}_t$ defined in equation (8).

7. Compute the sequence of production labor L_{Pt} using equation (6).

8. Compute the sequence of aggregate output growth $\frac{Y_{t+dt}}{Y_t}$ using equation (8).

9. With the paths of aggregate output growth and $\frac{Y_t}{w_t}$, obtain the implied path of wage growth $\frac{w_{d+dt}}{w_t}$.

10. Update the guesses for $\frac{w_{d+dt}}{w_t}$ and $\frac{Y_t}{w_t}$ using bisection and go back to step (a). Iterate until the guessed and implied paths converge.

I declare convergence when the difference between the three guessed and implied sequences is less than $\epsilon = 10^{-5}$ in absolute terms. I annualize wage growth before applying the convergence threshold.

[27] One could already simulate the entire two-dimensional distribution of quality and productivity gaps forward here. However, for the inner loop, it is sufficient to iterate on S_t.